

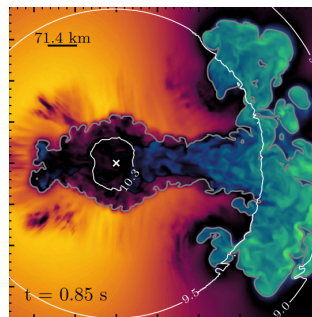
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Thermonuclear deflagration flames in type Ia supernovae

A 1D multiparametric study on the Phlegethon program



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Abstract

Type Ia supernovae, originating from the explosion of a white dwarf (WD) in a binary system, are standard candles used to measure cosmological distances and study dark energy. Understanding the physics of their explosion requires precise modeling of thermonuclear deflagration (deflagration) flames. This work presents a multiparametric study of these flame speeds using the stellar hydrodynamics code Phlegethon (Leidi et al., 2026). By extending existing models (Schwab et al., 2020; Timmes & Woosley, 1992), we look forward to establishing a new formulation for the laminar flame speed (flame speed) that accounts for the ambient density, chemical composition, and electron fraction of the medium. This extension paves the way for applying this to the study of deflagration flame speeds in stellar structures that evolve both spatially and temporally.

Keywords: Astrophysics, Type Ia supernova, Thermonuclear deflagration, Laminar flame speed, Nuclear reaction network, White dwarf, Phlegethon.

Résumé

Les supernovae de type Ia, issues de l'explosion d'une naine blanche en système binaire, sont des *chandelles standard* utilisées pour la mesure des distances cosmologiques et l'étude de l'énergie noire. Comprendre la physique de leur explosion nécessite une modélisation précise des flammes de déflagration thermonucléaire. Ce travail présente une étude multiparamétrique de la vitesse de ces flammes à l'aide du code d'hydrodynamique stellaire Phlegethon (Leidi et al., 2026). En étendant les modèles existants (Schwab et al., 2020; Timmes & Woosley, 1992), nous cherchons à établir une nouvelle formulation de la vitesse de flamme laminaire en prenant en compte la densité, la composition chimique et la fraction électronique du milieu. Cette extension pourrait permettre d'étudier la vitesse de déflagration de structures stellaires évolutives spatialement et temporellement.

Mots-clés : Astrophysique, Supernovae de type Ia, Déflagration thermonucléaire, Vitesse de flamme, Réseau thermonucléaire, Naine blanche, Phlegethon.

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1 Introduction

Type Ia supernovae are among the most important tools of modern cosmology, and their explosion mechanism, a thermonuclear deflagration in a degenerate carbon-oxygen white dwarf, remains an active area of research. A key ingredient in modeling these explosions is the speed at which the deflagration flame itself propagates, a quantity that depends on the local density, chemical composition, and electron fraction of the fuel, and that cannot currently be resolved directly in global explosion simulations.

This report presents the work carried out during a three-month *Projet de Recherche* (PRe) internship at the Heidelberg Institute for Theoretical Studies (HITS), within the Physics of Stellar Objects group, aimed at characterizing this laminar flame speed across a range of physical conditions using the stellar hydrodynamics code Phlegethon. Starting from no prior background in either astrophysics or large-scale numerical simulation, the internship combined learning the physical and numerical formalism underlying the code, running and validating a systematic parameter study, and developing the post-processing tools needed to extract a reliable flame speed from the resulting simulations, a task that turned out to be considerably less straightforward than initially expected.

The remainder of this report is organized as follows. Section 2 describes the working environment at HITS and the organization of the project. Section 3 introduces the astrophysical context and defines the scientific problem more precisely. Section 4 presents the scientific work itself: the mathematical and numerical formulation underlying Phlegethon (Section 4.1), the setup and preliminary numerical tests carried out before the main study (Section 4.2), the spectator-species methodology developed to vary the electron fraction (Section 4.3), the data post-processing pipeline built to extract flame speeds from simulation output (Section 4.4), the physical and numerical challenges encountered along the way (Section 4.5), the results obtained to date (Section 4.6), and a chronological account of the project (Section 4.7). Section 5 concludes the report and outlines the work that remains to be done.

2 Work Environment and Project Management

2.1 The Heidelberg Institute for Theoretical Studies (HITS) and the PSO Group

The work presented in this report was carried out at the **Heidelberg Institute for Theoretical Studies (HITS)**, a private, non-profit research institute dedicated to data-driven and computationally intensive natural sciences. HITS maintains strong academic ties with Heidelberg University while offering a state-of-the-art, highly autonomous research environment.

The research was conducted within the **Physics of Stellar Objects (PSO)** research group, led by **Prof. Dr. Friedrich K. Röpke**. The PSO group specializes in the numerical modeling of complex astrophysical processes, ranging from stellar evolution and binary star mergers (e.g., common envelope (CE) interactions) to Type Ia supernova explosions.

The group is housed across two distinct buildings within the HITS campus: the historic early 20th-century **Villa** (where part of the PSO team, including myself, was based) and the modern **Main Building**, which hosts the main group offices, meeting rooms, lunch room, and administration. Despite this physical distribution, cohesion is maintained through a highly collaborative work culture. Notably, PSO stands out as the most international research group at HITS. While many other groups predominantly speak German, PSO operates entirely in English, bringing together researchers from several continents with German native speakers accounting for less than a third of the team.

2.2 Work Infrastructure and Technical Ecosystem

2.2.1 Computational Facilities and Data Storage Strategy

Due to the multiphysical hydrodynamical and thermonuclear nature of Type Ia supernova simulations, access to high-performance computing (HPC) infrastructure is critical. Calculations were executed on **Genoa**, one of HITS's dedicated HPC clusters running under the SLURM workload manager. **Genoa** is a CPU cluster composed of 256 nodes each equipped with 256 logical cores (AMD EPYC 9174F, 16 physical cores per socket, 2 threads per core) and 384 GB of RAM. Contrary to other GPUS or CPU clusters available at HITS (like **Bergamo**, **Roma** or **Cascade**), it is not located at the HITS facility but is hosted in a building of the University, because of a cheaper electricity contract there. This also rules the way data is stored for the simulations and work we do at HITS.

Data storage is structured across two (soon to be, three) high-speed storage systems: fast (directly located into the HITS facility, used for the HITS-located clusters), and basement (located at the same place as the **Genoa** cluster) to store and use the data generated on the cluster without having to transfer it elsewhere.

- **Volume Demands:** Hydrodynamic runs generate massive outputs (density profiles, velocity fields, mass fractions across species). By the end of the internship, my datasets occupied several tens of Terabytes. Most people in the PSO group have datasets worth multiple hundreds of Terabytes.
- **Group-Wide Impact:** The PSO group alone accounts for over 60% of the total multi-Petabyte storage capacity of basement. This is the main reason why the IT team worked in close collaboration with the PSO group to manage better the upcoming storage system.
- **Infrastructure Challenges:** During my time at HITS, basement operated near critical capacity (~95% full). This required strict data hygiene (compressing raw outputs, removing redundant checkpoint files, and archiving) while the institute prepared to admit into active service the new storage system. The IT team held a lot of meetings and discussions with representatives of all the research groups to see how to best tune the settings and internal architecture of the new storage system to accommodate the diverse needs of the institute's research groups, especially in terms of bandwidth, data throughput and storage repartition. The cost of hardware, maintenance and electricity for storage space as well as clusters amounts to a significant portion of the institute's budget.

2.2.2 Software Engineering and Version Control

The core simulation software used throughout this project is **Phlegethon**, an in-house astrophysics code hosted on GitHub, complemented by GitLab repositories.

To implement the methodological updates required for my project, for example enabling true 1D cartesian geometry mode or implementing a dynamic OpenMP parallelization, a dedicated branch was created from the main repository by **Dr. Giovanni LEIDI**, in charge of **Phlegethon**. Development workflows were designed with code merging in mind, allowing optimizations proven during the project to be integrated back into Phlegethon's main codebase. The 1D option will most likely be merged into the main as an option for more efficient computing for people needing no more than one dimension, and the OpenMP parallelization is worked upon by other people and alumnis of the PSO group to be more efficient and usable on GPU clusters instead of only CPU ones like it is now.

2.3 Team Dynamics, Collaboration, and Communication

2.3.1 Supervisory Structure and Everyday Guidance

Project governance was organized around a dual tier of supervision:

1. **Academic Supervision:** Weekly progress reviews and strategic direction were provided directly by **Prof. Dr. Friedrich K. Röpke**, whose office in the Main Building remains accessible for any discussions at any point where he is available. This also tends to show once again the horizontal hierarchy within the group, which allows for a more collaborative and less-constrained approach to research and problem-solving.
2. **Daily Operations & Technical Mentorship:** Day-to-day progress was supported by three key members of the PSO group:

- **Kristián Vitovský (PhD Student):** He guided me with the initial onboarding, workspace configuration, and early familiarity with Phlegethon’s runtime parameters and execution debugging.
- **Dr. Giovanni Leidi (Postdoctoral Researcher):** As the primary developer of Phlegethon, Dr. Leidi assisted directly with code architecture, implementing customized code modifications (such as pure 1D geometries and enhanced data logging) on a dedicated git branch, as well as helping me understand the physics behind the results I got, whether those were expected or unplanned for.
- **Dr. Alexander Holas (Postdoctoral Researcher, PhD candidate during the internship):** He provided crucial expertise on the physical formulation of Type Ia supernovae. Dr. Holas guided the selection of nuclear networks, of initial conditions, and the physical approximations to ensure that numerical simplifications remained physically sound. He supported me all throughout the internship, and was the main point of contact for any questions regarding the project.

Support extended beyond office hours and contract end dates, with colleagues providing continuous technical and physical insights via Mattermost and in-person discussions.

2.3.2 Meeting Routines and Cross-Group Synergies

Research activities were structured around two fixed weekly meetings:

- **PSO Group Meeting (Wednesdays, 11:00 AM):** A joint session gathering the PSO team, members of the neighboring **SET** (Stellar Evolution Theory) group, and remote collaborators (including alumni now at institutions such as **Los Alamos National Laboratory**). Meetings combined operational updates (HPC allocations, upcoming conferences) with a 40 to 60 minutes long scientific presentation. These presentations provided a platform for thesis defenses, conference dry-runs, and peer feedback across diverse topics in stellar physics. It showed the place that multi-disciplinary and diversification has for the group, which is one of the main benefits of HITS not being only a university department focused on one topic only.
- **SLH Subgroup Meeting (Thursdays, 4:00 PM):** Dedicated specifically to Stellar Hydrodynamics (SLH), this afternoon meeting accommodated international remote participants with its time slot adjusted for time zone differences. Discussions were highly interactive, focusing on intermediate results, numerical anomalies, and technical blockers. Key technical breakthroughs, such as diagnosing dense-matrix bottlenecks in nuclear reaction networks ($O(N^3)$ scaling), identifying acoustic wave reflections at domain boundaries, and proposing dynamic OpenMP parallelization, originated directly from these sessions.

Working in fundamental computational astrophysics highlighted a fundamental difference between industrial engineering projects and academic research: project planning is inherently non-linear. Unlike classical engineering projects with pre-defined Gantt charts and rigid weekly milestones, numerical research involves constant adaptation. Unexpected physical phenomena, numerical instabilities, or stiff matrix convergence issues frequently generate new investigation branches. To

navigate this unpredictability, I managed priorities through dynamic task tracking and personal documentation files rather than trying to fit the work into a static initial timeline.

2.4 Integration, Working Conditions, and Scientific Culture

The working environment at HITS emphasizes both physical comfort and operational flexibility. Workstations in the open-plan offices feature highly ergonomic chairs and motor-driven sit-stand desks, equipped with dual-monitor setups and dedicated laptops connected to the institute's network via secure VPN and SSH tunnels.

From an organizational standpoint, HITS operates under a flexible office space policy (**flex-office**), allowing researchers to reserve any available desk within the PSO group's rooms via an internal booking system. In practice, however, team members naturally established preferred daily seats, striking a comfortable balance between organizational flexibility and a sense of routine.

Beyond formal meetings, scientific ideas and technical troubleshooting were routinely shared during coffee breaks and lunch hours. Digital collaboration was sustained through specialized Mattermost channels, there was one dedicated to every project being worked on by the people of the PSO group, as well as some dedicated to scientific literature tracking and competitor analysis, or informal chats.

Social integration was facilitated from the start through events such as the weekly "Mensa" evenings in Heidelberg, joint group hikes, and the annual PSO Summer Party. This inclusive culture fostered an ideal environment for scientific inquiry, international networking, and rapid professional as well as personal growth.

3 Astrophysical Background and Problem Definition

3.1 Cosmological and Astrophysical Significance of Type Ia Supernovae

Type Ia supernova (SN Ia) are among the most energetic transient phenomena in the Universe. Unraveling their physical explosion mechanisms is fundamental across multiple domains of modern physics:

1. **Cosmology and Dark Energy:** Due to their remarkable empirical uniformity, SN Ia serve as standardizable cosmological candles. Measurements of their luminosity distances led directly to the discovery of the accelerated expansion of the Universe (Perlmutter & others, 1999). However, improving the precision of dark energy constraints now requires reducing systematic uncertainties rooted in the unconstrained microphysics of the progenitor explosion.
2. **Galactic Chemical Evolution:** Thermonuclear explosions of compact remnants are the primary cosmic drivers for the synthesis and distribution of iron-peak elements (e.g., Fe, Ni, Cr) into the interstellar medium (Iwamoto et al., 2000).
3. **High Energy Density Physics (HEDP):** The extreme thermodynamic regimes reached inside exploding degenerate stellar matter, characterized by relativistic electron degeneracy, supersonic/subsonic burning fronts, and stiff nuclear reaction rates, share fundamental physical and mathematical aspects with some laboratory experiments, such as inertial confinement fusion (ICF).

Even if they play a crucial role for astrophysics, the exact physical initiation and propagation of the thermonuclear runaway remain subjects of active research. Simulating fully an exploding star is not computationally feasible, but we have ways of studying the phenomenon to feed models to the macro-scale simulations. This requires knowledge of the stellar objects we work upon, as well as the microphysical processes that govern the propagation of the thermonuclear flame front.

3.2 Stellar Progenitors and Microphysical Firefront Dynamics

3.2.1 Progenitor Systems and Nuclear Networks

Type Ia supernova (SN Ia) originate from carbon-oxygen (C-O) WD in binary systems that approach the Chandrasekhar mass limit via accretion or mergers. While oxygen-neon (O-Ne) WD systems also exist and undergo distinct evolutionary paths (Schwab et al., 2020), modeling C-O flames is central to understanding standard SN Ia explosions. From a computational perspective, simulating O-Ne mixtures in a physically coherent and comprehensive manner requires substantially larger nuclear reaction networks that struggle to tractably reach nuclear statistical equilibrium (NSE) within the current hydrodynamical and network capabilities of **Phlegethon** (Leidi et al., 2026).

3.2.2 The Multiscale Resolution Challenge and Deflagration Speed

In subsonic burning regimes (deflagration), the firefront propagates via thermal conduction mediated mainly by degenerate electrons. This microphysical structure introduces a severe multiscale challenge:

- The physical width of the laminar deflagration flame front (l_f) is dictated by microscopic transport coefficients and nuclear reaction timescales, spanning from a fraction of a micrometer to a few centimeters ($\sim 10^{-7}$ to 10 cm).
- The global radius of the parent WD is on the order of thousands of kilometers (10^8 cm).

Because global multi-dimensional hydrodynamic simulations operating at the stellar scale cannot resolve sub-centimeter flame structures, subgrid models or parameterized fits are required (Timmes & Woosley, 1992). The precise laminar flame speed v_l plays a decisive role in governing the transition to detonation and determining whether the star undergoes a complete explosion or a gravitational collapse (Holas & Röpke, 2026).

Determining accurate, localized laminar flame speeds v_l as a function of fuel density (ρ), chemical composition (X_i), and electron fraction (Y_e) is therefore an essential prerequisite for macro-scale astrophysical modeling. The current literature on the topic (Schwab et al., 2020; Timmes & Woosley, 1992) provides formulas that each take two parameters out of the three. The point of this work is to extend the parameter space exploration with all three parameters, and to provide a new fitting formula for the laminar flame speed $v_l = f(\rho, X_i, Y_e)$.

3.3 Numerical Framework and 1D Benchmarking in Phlegethon

To investigate these microphysical combustion processes, this study utilizes **Phlegethon** (Leidi et al., 2026), a modern computational astrophysics code designed for astrophysical fluid dynamics and nuclear burning.

The simulation setup is intentionally restricted to a 1D Cartesian geometry. This choice stems mainly from the fact that this work has been done by someone with no prior experience in astrophysics nor in numerical simulations, for a duration of only three months. However, the 1D approach still presents both scientific and practical advantages:

- **Simplicity and Unidimensional Front Analysis:** Planar 1D setups isolate the planar flame structure from geometry-induced curvature effects, allowing direct evaluation of thermal conduction and nuclear energy release across the firefront.
- **Framework Verification:** Running 1D benchmarks evaluates Phlegethon against established literature, verifying its numerical schemes, equation-of-state integrations, and nuclear reaction network solvers.
- **Profiling Framework Limits:** Operating in 1D allows testing the operational boundaries of the code, particularly identifying memory and runtime bottlenecks when scaling nuclear network sizes or approaching parameters for which the deflagration flame is hardly holding up.

3.4 Project Objectives and Key Deliverables

The primary goal of this project is to quantitatively model subsonic deflagration flames in degenerate stellar matter using Phlegethon. The work is structured around three core objectives:

1. **Laminar Flame Speed Formula:** Deriving a robust analytical fitting formula for the laminar flame speed v_l as a function of key physical input parameters:

$$v_l = f(\rho, X_i, Y_e) \quad (3.4.1)$$

where ρ represents the fuel density, X_i the initial chemical composition, and Y_e the electron fraction.

2. **Phlegethon Verification:** Validating Phlegethon's hydrodynamical and nuclear solvers by reproducing the results from the existing literature, and if the results differ, identifying the sources of discrepancy and address those by suggesting improvements.
3. **Code Profiling and Limitations:** Documenting the technical capabilities, numerical constraints, and potential avenues for future enhancements within the Phlegethon ecosystem. Produce documentation for the post-processing programs that have been developed for this project, and provide a clear roadmap for future work, should it come to use.

4 Scientific Work

4.1 Mathematical and Numerical Formulation

4.1.1 The Continuous Physical Model

Fluid Dynamics Equations for Stellar Plasmas

The governing equations describing the evolution of stellar plasma in Phlegethon are formulated as a hyperbolic-parabolic system of non-linear conservation laws with source terms (Leidi et al., 2026). In Cartesian geometry (or within generalized coordinate systems), the spatio-temporal evolution of the vector of conservative variables $U(x, t)$ under the action of the advective flux tensor $\mathbf{F}(U)$ and a vector of source terms $S(U)$ is written as (Euler, 1757; Leidi et al., 2026):

$$\frac{\partial U}{\partial t} + \nabla \cdot \mathbf{F}(U) = S(U) \quad (4.1.1)$$

The vector of conservative variables U collects the mass density ρ , the momentum density $\rho \mathbf{u}$ (where $\mathbf{u} = (u_x, u_y, u_z)^T$ is the fluid velocity vector), the total energy per unit volume E , and the partial mass densities of the various nuclear species ρX_k (Leidi et al., 2026):

$$U = \begin{pmatrix} \rho \\ \rho \mathbf{u} \\ E \\ \rho X_k \end{pmatrix} \quad (4.1.2)$$

The total energy per unit volume E combines the specific internal energy e_{int} and the macroscopic kinetic energy (Leidi et al., 2026):

$$E = \rho e_{\text{int}} + \frac{1}{2} \rho \|\mathbf{u}\|^2 \quad (4.1.3)$$

The mass fractions X_k associated with each isotope k strictly satisfy the mass conservation closure constraint $\sum_k X_k = 1$ (Leidi et al., 2026). The advective flux tensor $\mathbf{F}(U)$ represents the conservative transport of mass, momentum, and total energy through the fluid (Leidi et al., 2026):

$$\mathbf{F}(U) = \begin{pmatrix} \rho \mathbf{u} \\ \rho \mathbf{u} \otimes \mathbf{u} + P \mathbf{I} \\ (E + P) \mathbf{u} \\ \rho X_k \mathbf{u} \end{pmatrix} \quad (4.1.4)$$

where P denotes the total plasma pressure and $\mathbf{I}$ is the 3×3 identity matrix.

The source term $S(U)$ integrates the coupled contributions of gravity S_{grav} , diffusive thermal transport S_{diff} , and nuclear microphysics S_{nuc} (Leidi et al., 2026):

$$S(U) = S_{\text{grav}} + S_{\text{diff}} + S_{\text{nuc}} = \begin{pmatrix} 0 \\ \rho \mathbf{g} \\ \rho \mathbf{u} \cdot \mathbf{g} + \nabla \cdot (K \nabla T) + \rho \dot{\epsilon}_{\text{nuc}} \\ \rho \dot{\omega}_k \end{pmatrix} \quad (4.1.5)$$

where $\mathbf{g} = -\nabla \Phi$ is the gravitational acceleration field, K represents the total thermal conductivity, $\dot{\epsilon}_{\text{nuc}}$ is the specific nuclear energy release rate per unit time, and $\dot{\omega}_k$ is the net production or destruction rate of nuclear species k (Leidi et al., 2026).

Thermodynamic Closure (Equation of State)

The hydrodynamical system requires an equation of state (EoS) to close the system of equations by relating the pressure P and temperature T to the conservative variables $(\rho, e_{\text{int}}, X_k)$ (Leidi et al., 2026; Timmes & Arnett, 1999).

In Phlegethon, the Helmholtz EoS (HELMHOLTZ_EOS. In Phlegethon, physical modules and numerical solver variants are selected at compile time via define preprocessor flags specified in the code Makefile, denoted in MONOSPACE_FONT throughout this chapter) models the stellar medium as a superposition of three independent thermodynamic components (Leidi et al., 2026; Timmes & Arnett, 1999):

$$P(\rho, T, X_k) = P_{\text{ion}} + P_{\text{rad}} + P_{\text{elec/pos}} \quad (4.1.6)$$

1. **The ionic contribution (P_{ion}):** Ions are treated as a classical, fully ionized ideal gas obeying Maxwell-Boltzmann statistics (Leidi et al., 2026; Timmes & Arnett, 1999):

$$P_{\text{ion}} = n_{\text{ion}} k_B T = \frac{\rho}{|A| m_u} k_B T \quad (4.1.7)$$

where k_B is the Boltzmann constant, m_u is the atomic mass unit, and $|A| = \left(\sum_k \frac{X_k}{A_k} \right)^{-1}$ is the mean atomic mass of the mixture (Leidi et al., 2026).

2. **The radiation contribution (P_{rad}):** The radiation field is assumed to be in local thermodynamic equilibrium (LTE), following the Stefan-Boltzmann black body law (Leidi et al., 2026; Timmes & Arnett, 1999):

$$P_{\text{rad}} = \frac{1}{3} a T^4 \quad (4.1.8)$$

where $a = \frac{8\pi^5 k_B^4}{15c^3 h^3}$ is the radiation constant (Timmes & Arnett, 1999).

3. **The electron and positron contribution ($P_{\text{elec/pos}}$):** Electrons and positrons are modeled as an arbitrarily degenerate, relativistic Fermi-Dirac gas, including $e^- e^+$ pair production processes at very high temperatures (Leidi et al., 2026). Their thermodynamic quantities are computed via Fermi-Dirac integrals depending on the relative chemical potential $\eta = \frac{\mu}{k_B T}$ and the relativity parameter $\beta = \frac{k_B T}{m_e c^2}$ (Dirac, 1926; Fermi, 1926; Timmes & Arnett, 1999).

When the `USE_COULOMB_CORRECTIONS` option is enabled, electrostatic Coulomb correction terms are added to describe strongly coupled plasma regimes where the ion plasma coupling parameter $\Gamma = \frac{Z^2 e^2}{a_i k_B T} > 1$ (the ratio of electrostatic Coulomb potential energy between neighboring ions to their thermal kinetic energy) (Leidi et al., 2026; Salpeter, 1954).

Transport Terms and Nuclear Reactivity

Energy transport by thermal conduction and material transformations via nuclear reactions constitute two key microphysical source terms in stellar environments (Leidi et al., 2026).

- **Thermal conduction and opacities (`USE_TIMMES_KAPPA`):** The diffusive heat flux obeys Fourier's law $\mathbf{q}_{\text{cond}} = -K \nabla T$ (Fourier, 1822; Leidi et al., 2026). The total thermal conductivity K is related to the overall opacity of the medium κ through the radiative and conductive diffusion approximation (Leidi et al., 2026; Timmes & Arnett, 1999):

$$K = \frac{4acT^3}{3\kappa\rho} \quad (4.1.9)$$

The total opacity $\kappa = (\kappa_{\text{rad}}^{-1} + \kappa_{\text{cond}}^{-1})^{-1}$ combines the radiative opacity κ_{rad} (Thomson scattering, free-free transitions) and the electronic conductive opacity κ_{cond} , evaluated on the fly following the analytical prescriptions from Timmes and Leidi et al. (Leidi et al., 2026; Timmes & Arnett, 1999).

- **Nuclear kinetics and weak interaction rates (`USE_NUCLEAR_NETWORK`, `USE_LMP_WEAK_RATES`):** The temporal evolution of the mass fractions X_k driven by thermonuclear reactions and weak interactions is described by a coupled system of ordinary differential equation (ODE) (Arnett, 1996; Leidi et al., 2026):

$$\frac{dX_k}{dt} = \dot{\omega}_k(\rho, T, X_j) = \sum_j \lambda_{j \rightarrow k} X_j + \rho \sum_{j,l} \gamma_{jl \rightarrow k} X_j X_l + \dots \quad (4.1.10)$$

The network accounts for Coulomb barrier penetration with electron screening effects via `USE_ELECTRON_SCREENING` (Graboske et al., 1973) as well as weak interaction rates (electron captures and β decays) taken from the tabulated compilations of Langanke & Martínez-Pinedo (2000) via the `USE_LMP_WEAK_RATES` flag (Langanke & Martínez-Pinedo, 2000; Leidi et al., 2026).

4.1.2 Discretization and Numerical Solvers

The Finite Volume Framework

The continuous hyperbolic-parabolic system of conservation laws is discretized within the finite volume (FV) framework (Leidi et al., 2026). The spatial domain Ω is partitioned into non-overlapping control volumes or grid cells $K_i = [x_{i-\frac{1}{2}}, x_{i+\frac{1}{2}}]$ of volume $V_i = \Delta x_i$.

Unlike Finite Element methods that approximate solutions within continuous functional Sobolev spaces, the fundamental discrete variable in a FV scheme is not the pointwise value $U(x, t)$, but the cell-averaged conservative vector $|U|_{i(t)}$ defined over the Hilbert space $L^1(\Omega) \cap L^2(\Omega)$ (Leidi et al., 2026):

$$|U|_{i(t)} = \frac{1}{V_i} \int_{K_i} U(x, t) dx \quad (4.1.11)$$

Integrating the conservation law over the cell volume K_i and applying the **Green-Ostrogradsky divergence theorem** yields the exact semi-discrete integral formulation (Leidi et al., 2026):

$$\frac{d|U|_i}{dt} + \frac{1}{\Delta x_i} (F_{i+\frac{1}{2}} - F_{i-\frac{1}{2}}) = |S|_i \quad (4.1.12)$$

where $F_{i+\frac{1}{2}} \approx F(U(x_{i+\frac{1}{2}}, t))$ represents the numerical flux through the cell interface at $x_{i+\frac{1}{2}}$.

This formulation guarantees exact numerical conservation at machine precision: the flux leaving cell K_i through face $i + \frac{1}{2}$ is identically equal to the flux entering cell K_{i+1} ($F_{i+\frac{1}{2}} \equiv F_{(i+1)-\frac{1}{2}}$). Furthermore, according to the Lax-Wendroff theorem, if a conservative numerical scheme converges under grid refinement $((\Delta x, \Delta t) \rightarrow (0, 0))$, its limit is guaranteed to be a weak solution of the continuous Euler equations, ensuring the exact capture of shock jump conditions (Rankine-Hugoniot relations) (Hugoniot, 1887; Lax & Wendroff, 1960; Rankine, 1870).

Spatial Discretization: Reconstruction and Riemann Solver

To compute high-order interface fluxes without generating parasitical numerical oscillations near steep gradients or shocks, Phlegethon combines a high-order spatial reconstruction scheme with an approximate Riemann solver (Leidi et al., 2026).

- **Fifth-Order Spatial Reconstruction (LIM5TH_REC):** Interface states $U_{i+\frac{1}{2}}^L$ and $U_{i+\frac{1}{2}}^R$ are reconstructed from the cell averages $|U|_i$ using the 5th-order Monotonicity-Preserving (5th-order Monotonicity-Preserving scheme (MP5)) scheme (Leidi et al., 2026; Suresh & Huynh, 1997). The 5th-order Monotonicity-Preserving scheme (MP5) algorithm interpolates interface values with 5th-order spatial accuracy in smooth regions while applying a multi-stage limiter that monitors local cell curvature to preserve monotonicity near discontinuities without clipping physical extrema (Suresh & Huynh, 1997). Third-Order reconstruction is also an available option. In the case of this 1D study, it was needed to add ghost cells at the domain boundaries to ensure proper reconstruction in the directions where only one cell is present (3 ghost cells for the fifth order reconstruction).
- **HLLC Riemann Solver (HLLC_FLUX):** The local Riemann problem at interface $x_{i+\frac{1}{2}}$ defined by the state pair $(U_{i+\frac{1}{2}}^L, U_{i+\frac{1}{2}}^R)$ is resolved using the HLLC (Harten-Lax-van Leer-Contact) approximate solver (Leidi et al., 2026; Toro et al., 1994). The HLLC solver restores the middle contact discontinuity wave S^* omitted by the standard HLL solver, resolving a 3-wave, 4-state structure (S_L, S^*, S_R) (Toro et al., 1994):

$$F_{i+\frac{1}{2}}^{\text{HLLC}} = \begin{cases} F(U^L) & \text{if } 0 \leq S_L \\ F(U^L) + S_L(U^{*L} - U^L) & \text{if } S_L \leq 0 \leq S^* \\ F(U^R) + S_R(U^{*R} - U^R) & \text{if } S^* \leq 0 \leq S_R \\ F(U^R) & \text{if } 0 \geq S_R \end{cases} \quad (4.1.13)$$

- **Low-Mach Correction (Mach_cutoff_make=1.0e-4_rp):** In highly subsonic flow regimes (*Mll1*), standard Godunov-type fluxes suffer from excessive numerical dissipation. Setting the low-Mach cutoff parameter scales the wave speed dissipation to maintain accuracy and prevent damping of mild convective modes in stellar matter (Leidi et al., 2026).

Temporal Integration: SSP-RK3

The system of ordinary differential equations resulting from spatial discretization, $\frac{d|U|}{dt} = \mathcal{L}(|U|)$, is integrated in time using the 3rd-order Strong Stability Preserving Runge-Kutta Order 3 (SSP-RK3) scheme (RK3_STEPPER) (Leidi et al., 2026; Shu & Osher, 1988).

The Strong Stability Preserving Runge-Kutta Order 3 (SSP-RK3) algorithm advances the solution from U^n to U^{n+1} through three convex combinations of Forward Euler steps, preserving non-linear stability and TVD (Total Variation Diminishing) properties (Shu & Osher, 1988):

$$\begin{aligned} U^{(1)} &= U^n + \Delta t \mathcal{L}(U^n) \\ U^{(2)} &= \frac{3}{4}U^n + \frac{1}{4}U^{(1)} + \frac{1}{4}\Delta t \mathcal{L}(U^{(1)}) \\ U^{n+1} &= \frac{1}{3}U^n + \frac{2}{3}U^{(2)} + \frac{2}{3}\Delta t \mathcal{L}(U^{(2)}) \end{aligned} \quad (4.1.14)$$

The explicit time step Δt is updated dynamically (VARIABLE_TIMESTEP) based on the Courant-Friedrichs-Lewy (CFL) condition (cfl_make=0.8_rp) (Leidi et al., 2026):

$$\Delta t_{\text{hydro}} = C_{\text{CFL}} \cdot \min_i \left(\frac{\Delta x_i}{\max(|\mathbf{u}_i|) + c_{s,i}} \right) \quad (4.1.15)$$

with $C_{\text{CFL}} = 0.8$ and c_s the local sound speed. The value of the CFL was sometimes reduced down to 0.3 in order to ensure stability while testing new parameters or more instable flame fronts.

Microphysics Acceleration and Stiff Operator Treatment

- **FastEOS Acceleration (USE_FASTEOS):** Calling the Helmholtz equation of state (EoS) table repeatedly during the iterative Riemann flux evaluations at every cell interface is computationally costly, hence it is prohibitive. The FastEOS algorithm evaluates two auxiliary thermodynamic indices at cell centers (Leidi et al., 2026):

$$\gamma_e = \frac{P}{\rho e_{\text{int}}} + 1 \quad \text{and} \quad \gamma_c = \frac{\rho c_s^2}{P} \quad (4.1.16)$$

These indices are interpolated to cell interfaces $x_{i+\frac{1}{2}}$, enabling direct calculation of interface pressures and sound speeds without iterative Newton-Raphson inversions, reducing EoS evaluation by a factor of up to 6 (Leidi et al., 2026).

- **Thermal Diffusion via Super-Time-Stepping (THERMAL_DIFFUSION_STS):** Explicit time-stepping for parabolic heat conduction is constrained by the restrictive stability limit $\Delta t_{\text{diff}} \propto \Delta x^2$. Phlegethon circumvents this bottleneck using the second-order Runge-Kutta-Legendre Super-Time-Stepping (STS) scheme (Leidi et al., 2026; Meyer et al., 2012). The STS algorithm executes a sub-cycle of s stabilized explicit Tchebychev/Legendre stages to match the hydrodynamical time step (Meyer et al., 2012):

$$\Delta t_{\text{STS}} = \Delta t_{\text{hydro}} = \sum_{j=1}^s \tau_j \approx \frac{s^2 + 2s - 2}{2} \Delta t_{\text{diff}} \quad (4.1.17)$$

lifting the parabolic constraint while maintaining explicit algorithm efficiency (Leidi et al., 2026; Meyer et al., 2012).

- **Implicit Nuclear Network Integration (NUCLEAR_NETWORK_BE):** Due to the extreme stiffness of nuclear reaction timescales, the thermonuclear network is integrated using a 1st-order Backward Euler (Backward Euler (BE)) implicit scheme (Leidi et al., 2026; Timmes, 1999):

$$X_k^{n+1} - X_k^n = \Delta t \dot{\omega}_k(\rho^{n+1}, T^{n+1}, X_j^{n+1}) \quad (4.1.18)$$

This non-linear system is solved locally in each cell using a multi-dimensional Newton-Raphson solver backed by optimized linear algebra libraries (Leidi et al., 2026; Timmes, 1999).

4.1.3 Software Implementation and HPC Architecture

Code Architecture and Execution Flow

The Phlegethon code is implemented in modern, modular Fortran (Fortran 2008/2018 standards) optimized for massively parallel high-performance computing (high-performance computing (HPC)) environments (Leidi et al., 2026). The framework enforces a strict segregation of responsibilities between compile-time numerical definitions, hardcoded physical simulation parameters, and dynamic execution management on HPC clusters.

As illustrated in Figure 4.1.1, the simulation setup is structured across three distinct layers:

- **Numerical & Discretization Parameters (Makefile):** Spatial grid resolution (nx1, nx2, nx3), ghost cell counts, nuclear reaction network size (number of isotopes and reactions), and high-order numerical scheme flags (OPTS) are defined at compile time. This allows the compiler to generate fully optimized static array allocations and remove inactive physical modules.
- **Physical & Boundary Configuration (app.F90):** Physical domain bounds (e.g., x2l and x2u defining lower/upper physical limits in centimeters), initial profiles, and physical boundary conditions are hardcoded within the primary driver module app.F90.
- **HPC Resource Management (run.genoa):** High-level cluster execution parameters are isolated from code compilation. The run.genoa submission script defines the hardware footprint on the cluster, specifying the requested number of compute nodes, CPU cores, and parallel MPI tasks, and issues the launch command (e.g., mpirun).

The algorithm follows a strict sequence during each global time step Δt :

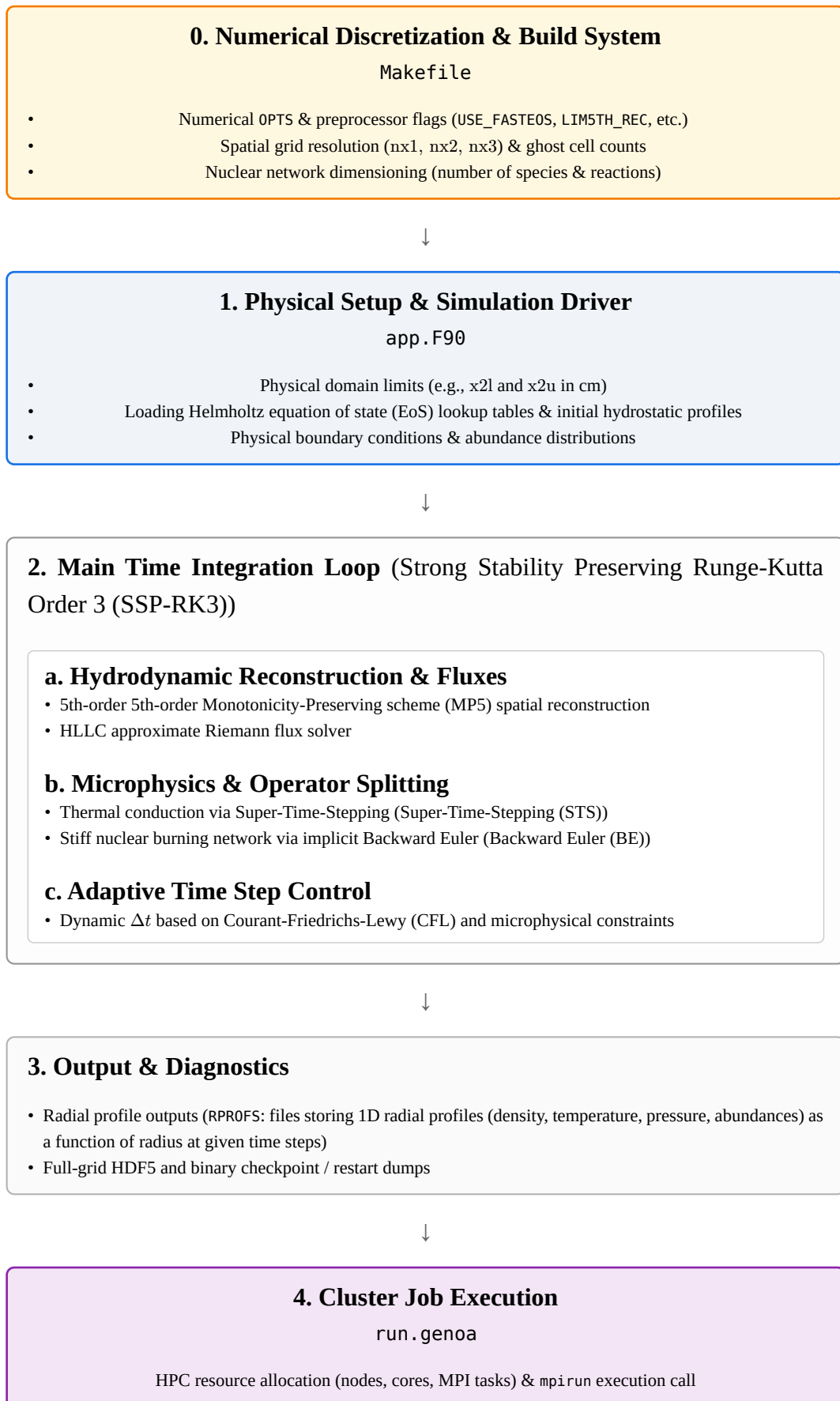


Figure 4.1.1: Schematic overview of the Phlegethon software architecture, separating numerical build setup (Makefile), physical configuration (app.F90), core integration steps, and HPC execution control (run.genoa).

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1. **Domain Setup & Initialization:** The driver parses execution options, initializes grid structures, loads pre-computed tabular data (e.g., Helmholtz equation of state (EoS) tables), and applies physical boundary conditions.
2. **Explicit Hydrodynamic Update:** The spatial reconstruction 5th-order Monotonicity-Preserving scheme and interface flux calculations (HLLC) are evaluated across all spatial cells. Time advancement is performed via the Strong Stability Preserving Runge-Kutta Order 3 integrator.
3. **Microphysics & Source Term Integration:** Non-advective source terms are integrated using operator splitting. Thermal conduction is advanced via STS, while thermonuclear reaction networks are solved implicitly using BE with a localized Newton-Raphson solver.
4. **Adaptive Time Step Control:** The dynamic time step Δt for the subsequent iteration is evaluated based on the minimum constrained timescale across all physical processes (CFL condition, diffusion limit, and nuclear burning rates).

As detailed in Figure 4.1.1, the numerical strategy relies on a operator-splitting approach to decouple the explicit hydrodynamic transport from the stiff microphysical source terms. During each global time step Δt , the hydrodynamic fluxes are integrated explicitly using the Strong Stability Preserving Runge-Kutta Order 3 (SSP-RK3) scheme, whereas thermal conduction (STS) and stiff nuclear kinetics (Backward Euler) are solved as sub-cycled or implicit steps.

To maintain global stability and accuracy without triggering numerical artifacts, the dynamic time step Δt is adaptively constrained at the end of each iteration by taking the minimum timescale across all physical modules:

$$\Delta t = \eta \cdot \min(\Delta t_{\text{CFL}}, \Delta t_{\text{diff}}, \Delta t_{\text{nuc}}) \quad (4.1.19)$$

where $\eta < 1$ is a safety factor, Δt_{CFL} represents the hydrodynamic Courant-Friedrichs-Lewy condition, Δt_{diff} the thermal diffusion limit, and Δt_{nuc} the nuclear burning timescale dictated by species abundance variations.

Modular Compilation and Makefile Configuration

Physical options and numerical schemes in Phlegethon are toggled at compile time using preprocessor macro flags defined within the `Makefile`. By hardcoding the numerical grid resolution (`nx1`, `nx2`, `nx3`) and array dimensions directly into the build configuration, compiler-level vectorization and memory layouts are optimized for the targeted HPC hardware.

The primary compile-time preprocessor options utilized for the physical modeling presented in this study are summarized in Table 4.1.1.

Key parameter blocks defined in `Makefile` include:

- **Grid Resolution and Extent:** Configuration of the spatial domain boundaries, cell counts (`nx1`, `nx2`, `nx3`), and coordinate geometry.
- **Courant Parameter:** Setting `cfl_make=0.8_rp` ensures strict compliance with the CFL hydrodynamic stability criterion.

Makefile Flag	Preprocessor	Physical / Numerical Module	Description & Reference
USE_FASTEOS		Fast EOS Interpolation	Accelerates thermodynamic lookup calls (Leidi et al., 2026)
HELMHOLTZ_EOS		Helmholtz Equation of State	Tabulated degenerate EoS (Timmes & Arnett, 1999)
USE_TIMMES_KAPPA		Conductive & Radiative Opacities	Timmes opacity evaluations (Timmes & Arnett, 1999)
USE_NUCLEAR_NETWORK		Thermonuclear Network	Coupled nuclear kinetics solver (Timmes, 1999)
USE_LMP_WEAK_RATES		Weak Interaction Rates	Langanke & Martínez-Pinedo rates (Langanke & Martínez-Pinedo, 2000)
USE_ELECTRON_SCREENING		Coulomb Electron Screening	Screening corrections (Graboske et al., 1973)
LIM5TH_REC		5th-Order MP5 Reconstruction	Monotonicity-preserving spatial scheme (Suresh & Huynh, 1997)
HLLC_FLUX		HLLC Riemann Solver	3-wave approximate Riemann flux (Toro et al., 1994)
RK3_STEPPER		SSP-RK3 Integrator	3rd-order TVD time integration (Shu & Osher, 1988)
THERMAL_DIFFUSION_STS		Super-Time-Stepping (STS)	Accelerated explicit heat diffusion (Meyer et al., 2012)
NUCLEAR_NETWORK_BE		Implicit Backward Euler	Stiff nuclear network integrator (Timmes, 1999)

Table 4.1.1: Key compile-time preprocessor flags configured in the Phlegethon Makefile for stellar hydrodynamics simulations.

- **Low-Mach Numerical Dissipation Cutoff:** Setting `Mach_cutoff_make=1.0e-4_rp` scales wave-dissipation terms in subsonic convective flows to minimize artificial damping.
- **Physical Boundary Conditions:** Definition of reflective, outflow, or hydrostatic boundary states at domain boundaries.
- **I/O and Diagnostics Controls:** Frequency of binary/HDF5 data dumps, checkpoint restarts, and integrated energy/mass conservation logging.

Runtime Parameters and Run Configuration (`run.genoa`)

While physical solvers and numerical schemes are compiled directly into the executable, the HPC-control parameters are controlled via the `run.genoa` input configuration file, in which are defined things such as the number of nodes of the cluster you ask for, the number of cores per node, the number of MPI tasks, and the command to launch the executable (e.g., `mpirun`). This separation allows for flexible execution on different HPC clusters without modifying the code.

4.2 Setup and Preliminary Numerical Tests

4.2.1 Simulation Box Configuration and Discretization

The flame propagates along the x_2 direction of the computational domain, with the transverse directions x_1 and x_3 left inactive in this 1D setup. The reference configuration uses $x_{2l} = 0$ and $x_{2u} = 4.0 \times 10^{-3}$ cm, discretized with $\text{nx2} = 2048$ cells, giving a cell width $\Delta x = x_{2u} / \text{nx2} \approx 1.95 \times 10^{-6}$ cm.

The spatial resolution was set following a criterion of at least ten grid cells per expected flame width, adopted on a colleague’s recommendation. For the reference case ($\rho = 3 \times 10^9$ g/cm³, $X(\text{C12}) = 0.5$), this resolution corresponds to approximately 13-14 cells per flame width.

Automatic box sizing. Because the flame width and propagation speed vary by orders of magnitude across the explored parameter space (Section 4.2.4), a single fixed box configuration is unsuitable for the full phase-space sweep. The `generate_simulations.py` script instead sizes every simulation individually from the flame width and speed predicted by (Timmes & Woosley, 1992):

$$x_{2u} = N_{\text{widths}} \times \ell_{\text{Timmes}}, \quad \text{nx2} = N_{\text{widths}} \times N_{\text{cells/width}} \quad (4.2.1)$$

with $N_{\text{widths}} = 300$ and $N_{\text{cells/width}} = 10$, the resulting cell count being rounded up to the next power of two for compatibility with the domain decomposition. Because every simulation is sized using the same $N_{\text{cells/width}}$, all simulations share the same spatial resolution relative to the local flame width, independent of where they sit in the parameter space. In practice, this rounding consistently lands on $\text{nx2} = 2048$ across the explored parameter space (see the computational-cost discussion below). The simulation end time is set so that the flame crosses the box exactly once,

$$t_{\text{max}} = x_{2u} / v_{\text{cond, Timmes}}. \quad (4.2.2)$$

The factor of 300 flame widths was chosen to leave additional margin for the transient to relax before the flame reaches the domain boundary (Section 4.5).

An analytical argument explains why nx2 stays close to a single value across the whole parameter space, even though it is formally recomputed from Equation (4.2.1) for every simulation. Because the flame speed and flame width predicted by (Timmes & Woosley, 1992) vary in inverse proportion to one another across (ρ, X_i) , a faster-burning mixture having a correspondingly narrower flame, so the product $N_{\text{widths}} \times N_{\text{cells/width}}$ that sets nx2 varies comparatively little over the explored range, and its nearest power of two consistently rounds to $\text{nx2} = 2048$. A fast, thin flame is thus resolved at the same effective grid resolution as a slow, thick flame integrated for a correspondingly longer physical time, keeping the computational cost per simulation roughly constant across the phase space.

Ignition profile. The flame is seeded at $t = 0$ with a hyperbolic tangent transition between a pre-burned (“ash”) region and the cold background,

$$T(x) = T_b + T \times 0.5(1 - \tanh((x - x_{2u}) / \Delta x)) \quad (4.2.3)$$

where x_t denotes the fraction of the box length initially set to the ignition temperature T , i.e. the extent of the pre-ignited “ash” region, and deltax sets the sharpness of the transition to the cold background at $T_b = 0.01T = 10^8$ K. A convergence study over x_t and deltax (Section 4.2.4) led to fixing $\text{deltax} = x_t / 2$, so that the transition width scales with the ignited-region size rather than being an independent free parameter. The absolute length of ignited material, $L_{\text{ign}} = x_t \times x_{2u}$, is kept constant across all simulations by rescaling x_t whenever x_{2u} changes with the target parameters, keeping the initial energy injected into the system comparable across the parameter space. The peak ignition temperature was initially set to $T = 10^{10}$ K (see Section 4.2.4 for a complication this raised).

(Schwab et al., 2020) (Appendix A) show that the flame speed is insensitive to the precise values of x_t , deltax , and T_b , provided the ignited region is neither so small that the flame extinguishes nor so large that it produces a prolonged, decelerating transient; after a few flame widths of propagation, the memory of the initial condition is effectively erased. This motivates fixing the ignition parameters once rather than tuning them per simulation, at the cost of the transient-length sensitivity discussed in Section 4.2.4.

4.2.2 Nuclear Reaction Network Selection

The nuclear network used by Phlegethon was progressively extended over the course of the internship:

- **7 species** (he4, c12, o16, ne20, mg24, si28, s32; ~ 20 reactions): a minimal α -chain network, used as a stable functional baseline.
- **15 species**: adds n14, ne22, ar36, ca40, ti44, cr48, fe52, ni56, completing the α -chain up to ni56. Functional, and used for the majority of the preliminary tests reported in this section.
- **35 species** (current production network): extends the 15-species set with non- α isotopes (n, p, na22, na23, mg25, mg26, al27, si29, si30, p31, and others). Flame speeds obtained with this network are slightly below those of (Timmes & Woosley, 1992), an expected consequence of the smaller network size (see below) rather than an indication of a numerical problem.
- **56 species**: currently non-functional in Phlegethon. The most likely cause is that the nuclear network solver relies on dense matrices for its linear algebra; even though these matrices are mostly filled with zeros for a network of this size, dense storage and factorization still scale as $O(n^3)$ in the number of species, making the 56-species case prohibitively expensive and triggering the errors observed. A sparse-matrix formulation is a candidate fix, discussed further as future work (Section 4.5); this has not been resolved at the time of writing.

For reference, (Timmes & Woosley, 1992) use a 130-isotope network (their Table 1, network 5), while (Schwab et al., 2020) use an adaptive nuclear reaction network (network) of ~ 495 isotopes, noting that more than 200 isotopes are required for their results to converge. Table 5 of (Timmes & Woosley, 1992) shows the conductive flame speed increasing systematically with network size (9, 19, 33, 83, and 130 isotopes give 54.4, 71.9, 79.1, 87.2, and 87.5 km/s respectively, for an O+Ne+Mg mixture at $\rho_9 = 10$), which would naively suggest that the present 35-species results

should underestimate the true, large-network flame speed. In practice, this is not clear-cut: the 35-species network already includes most of the highest-energy-yield reactions, but omits others that would reduce the net energy release if included, so whether the resulting speed is biased high or low is not obvious in general and may depend on the local conditions, density in particular. This is an important caveat when interpreting the validation against (Timmes & Woosley, 1992) in Section 4.6, and part of the motivation for tracking the overall trend across the parameter space, via the polynomial fit of Section 4.4.3, rather than relying solely on the absolute agreement of any single simulation.

4.2.3 OpenMP Parallel Scalability

Simulations run on the Genoa cluster’s nodes, each equipped with 256 logical cores (AMD EPYC 9174F, 16 physical cores per socket, 2 threads per core) and 384 GB of RAM. Table 4.2.1 summarizes a scalability test using the 15-species network, run for 10 000 timesteps at $nx2 = 8192$, comparing pure HPC-style MPI parallelism against hybrid MPI+OpenMP configurations.

MPI ranks	OMP_NUM_THREADS	wall-clock time (s, $\times 10^2$)	Total duration
64	1	0.575	1min 22s
64	2	0.323-0.564 (variable)	N/A
64	4	0.554	1min 20s
64	8	0.977	2min 18s
64	16	1.320	3min 06s

Table 4.2.2: OpenMP scalability test, 15-species network, 10 000 timesteps, $nx2 = 8192$.

OpenMP provides no measurable benefit at this problem size and network: OMP_NUM_THREADS=2 gives highly variable timings, consistent with memory contention rather than genuine parallel speed-up, and performance degrades monotonically beyond two threads (Table 4.2.2). A representative radial-profile overview from one of these runs is given in Appendix A. It remains possible that larger networks (35 or 56 species), for which the nuclear network integration cost is expected to weigh more heavily in the total runtime, would better justify hybrid parallelization; this has not yet been confirmed. Despite the marginal and inconsistent benefit measured in this test, production runs use OMP_NUM_THREADS=2 alongside MPI rather than pure MPI.

4.2.4 Convergence and Sensitivity Studies

Four numerical or initial-condition parameters were varied to test their effect on the extracted flame speed: grid resolution, box size, ignition temperature, and ignition spot size.

Grid resolution and box size. The ten-cells-per-flame-width criterion and the 300-flame-width box length (Section 4.2.1) were adopted following a colleague’s recommendation and a qualitative expectation that a longer box improves convergence margin, respectively, rather than from a dedicated convergence study; a systematic test on extreme phase-space cases is left for future work. Whether the chosen margin was ultimately sufficient is discussed in Section 4.5.

Ignition temperature. The initial choice of $T = 10^{10}$ K for the ignited zone was found to be problematic at lower densities: at $\rho = 10^8$ g/cm³, this temperature exceeds the upper validity range of the JINA REACLIB reaction rate library (Cyburt et al., 2010), triggering spurious photo-disintegration ($\dot{\epsilon}_{\text{nuc}} < 0$) within the ignition zone itself, as illustrated in Figure 4.2.2. We can also qualitatively notice that a higher ignition temperature will pour additional energy into the system, hence adding amplitude and duration to the transient.

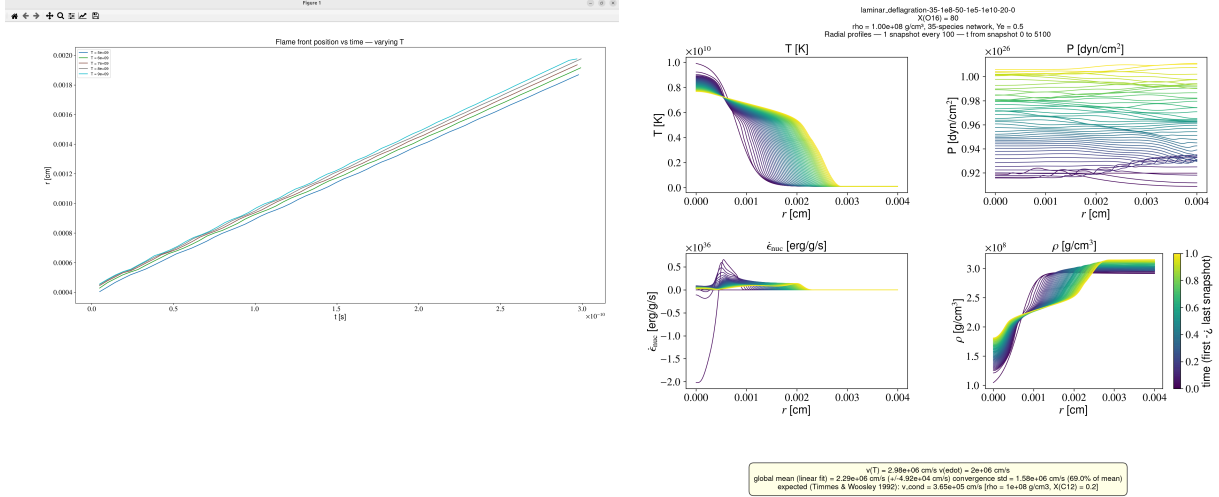


Figure 4.2.2: Left: comparison graph of the flame front position for different ignition temperatures T . The gap between the curves is smaller than our grid resolution for these simulations. Right: negative $\dot{\epsilon}_{\text{nuc}}$ in the ignition zone at $\rho = 10^8$ g/cm³, $T = 10^{10}$ K: this run did not produce a self-sustained flame.

A lower ignition temperature ($T = 10^9$ K) was tested to address this; a definitive value had not yet been settled at the time this investigation was carried out (see Section 4.5 for the eventual resolution of this issue).

Ignition spot size. With the 15-species network, $x_t = 0.004$ successfully ignites a self-sustained flame, while $x_t = 0.002$ fails to do so: in this case the perturbation simply lacks enough energy to trigger a self-sustained reaction and decays by pure diffusion, as illustrated in Figure 4.2.3, rather than being affected by the photodisintegration issue discussed above. The 35-species network, with more nuclear energy available, is expected to ignite at a lower threshold, though this was not separately quantified. A separate unfavorable test case ($\rho = 10^8$ g/cm³, $X(\text{C12}) = 0.2$, $x_t = 0.2$) did show a visible ignition attempt but an unclear, disrupted front, linked instead to the photodisintegration issue described above. As described in Section 4.2.1, the absolute ignited length L_{ign} was ultimately fixed across all simulations (via the x_t rescaling described there) rather than tuned per parameter point.

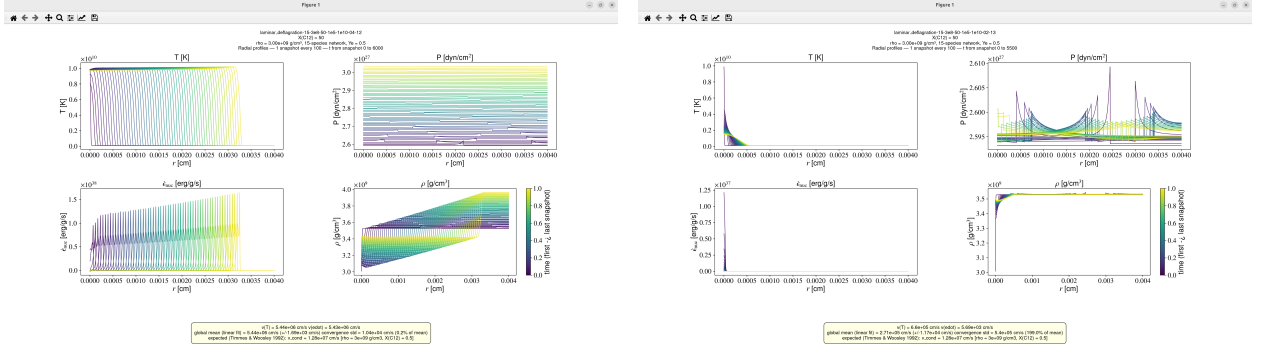


Figure 4.2.3: Ignition outcome with the 15-species network: $x_t = 0.004$ (left) successfully ignites a self-sustained flame; $x_t = 0.002$ (right) fails to ignite and decays by pure diffusion.

Common outcome. Across all four parameters, none was found to change the converged, late-time flame speed: only the duration and amplitude of the preceding transient were affected. This behavior, initial and numerical conditions altering the approach to a solution without altering the solution itself, is consistent with the flame speed being a stable physical attractor of the underlying reaction-diffusion dynamics, discussed further in Section 4.5.

4.2.5 Spurious Acoustic Wave Reflections

Oscillations superposed on the instantaneous flame speed, visible in early diagnostic plots, were at first attributed to noise on the data. However, due to the consistent nature of the oscillations, the hypothesis was made that these came from acoustic waves reflecting back and forth across the simulation domain. The boundary conditions active in the current configuration (defined in the Makefile) are reflective at both ends of x_2 and periodic (inactive) in x_1 and x_3 . To test this theory, a comparison study was made in Figure 4.2.4 to check if the frequency (and period) of the oscillations grew with $x_2 u$, the length of the box, because that is what would be expected of constant-speed waves bouncing back and forth in a medium of changing length.

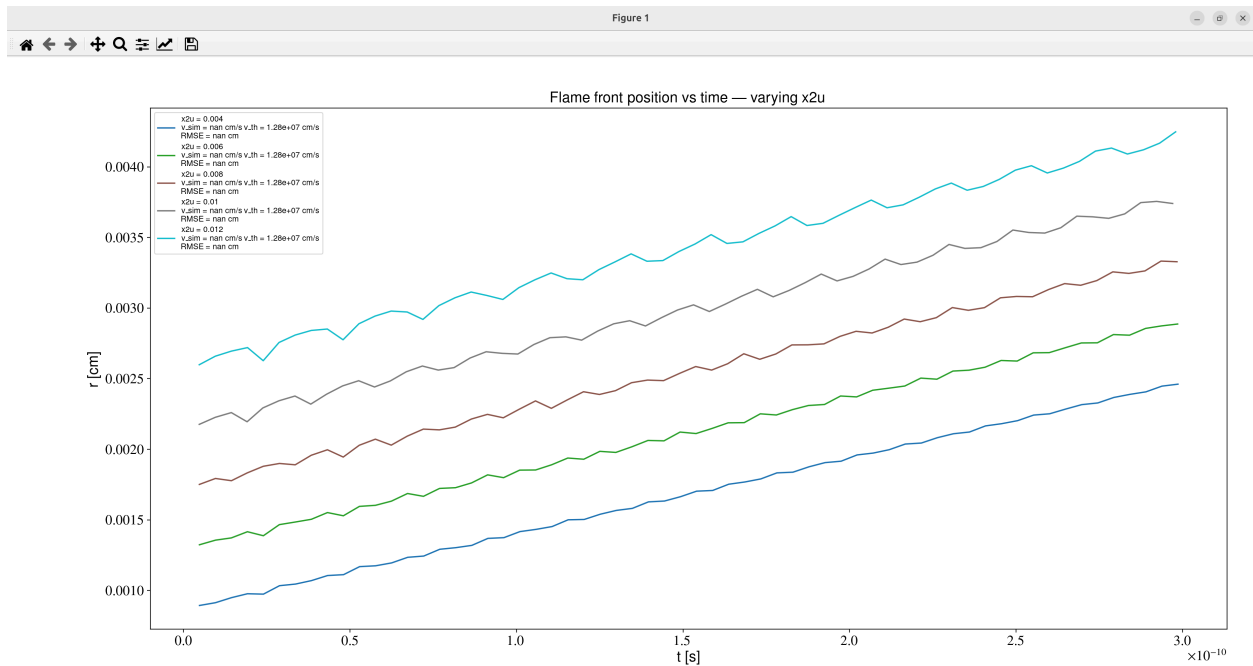


Figure 4.2.4: Position of the flame front as a function of time, with varying box lengths $x2u$.

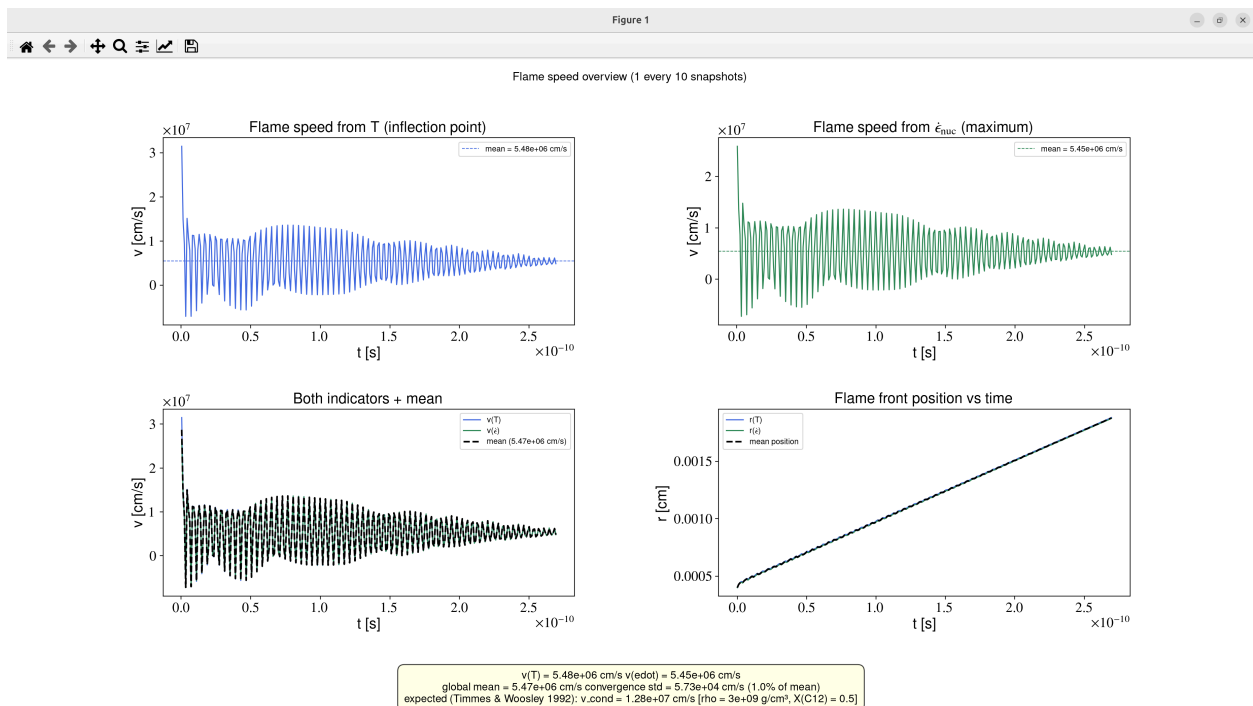


Figure 4.2.5: Instantaneous flame speed (both indicators and their mean) over a full run, showing the reflected wave's oscillation progressively damping out as the transient relaxes.

The physical origin of these waves is discussed further in Section 4.5, along with their consequences for flame-speed extraction: briefly, the artificial energy injected into the domain at ignition also seeds a compression wave. As this wave bounces back and forth across the box, it locally perturbs the density of the medium at the flame front; since the flame speed itself depends on

the local density of the stellar matter, each pass of the wave modulates the instantaneous speed measurement, hence even more noise in the computed speed value.

Alternative boundary conditions were tested in an attempt to remove these reflections at the source: inflow and outflow conditions at x_{2l} and x_{2u} respectively, as well as far-field conditions. Each alternative introduced its own artifacts and reproduced the deflagration physics noticeably less faithfully than the simple reflective (bounce-back) conditions, which were therefore retained despite the reflections they permit.

The mitigation adopted at this stage of the project was to add a fit to the flame front position with a low-order (third-degree) polynomial in addition to differencing consecutive raw snapshot positions, as this new method yields an analytically smooth derivative for the instantaneous speed. As the parameter space exploration progressed, the noise on the signal became a major obstacle to reliable flame-speed extraction, discussed extensively in Section 4.5.

4.3 Spectator Species Methodology for Variable Y_e

4.3.1 Motivation

The electron fraction Y_e of the fuel is not universal across progenitor white dwarfs, nor necessarily uniform within a single one, and it directly affects the propagation of the deflagration flame (Schwab et al., 2020). Two distinct physical drivers are relevant here. First, Y_e varies from one progenitor to another as a consequence of the initial metallicity of the main sequence star: CNO-cycle products converted to neutron-rich ^{22}Ne during helium burning set a metallicity-dependent neutron excess in the resulting C-O core (Timmes et al., 2003). Second, within a given white dwarf, Y_e can be non-uniform as a result of the star's specific burning history, stratified across different layers (Jones et al., 2013). Reproducing this dependence numerically requires a way to vary Y_e independently of the C-O composition already explored in Section 4.2, while staying physically sound, and without introducing a physically inconsistent nuclear species into the burning process itself.

4.3.2 The Ne40 Spectator Species

No naturally abundant isotope provides the exact Y_e shift required without substantially perturbing the C-O fuel composition. The approach adopted instead introduces Ne40 ($Z=10$, $A=40$) as an inert spectator species: an unrealistic but numerically convenient isotope with $Z/A = 0.25$, added to the fuel composition without taking part in the reaction network.

For a mixture of C12, O16 (each with $Z/A = 0.5$) and a mass fraction $X_s(\text{Ne40})$ of the spectator species,

$$Y_e = 0.5(1 - X_s(\text{Ne40})) + 0.25X_s(\text{Ne40}) \quad (4.3.1)$$

which rearranges to

$$X_s(\text{Ne40}) = (0.5 - Y_e)/0.25 \quad (4.3.2)$$

The remaining mass fraction is then distributed between C12 and O16 according to the target C/O ratio,

$$\begin{aligned} X_s(\text{C12}) &= \text{ratio}_{\text{C/O}} \times (1 - X_s(\text{Ne40})), \\ X_s(\text{O16}) &= (1 - \text{ratio}_{\text{C/O}}) \times (1 - X_s(\text{Ne40})) \end{aligned} \quad (4.3.3)$$

so that Equation (4.3.2) and Equation (4.3.3) together fully specify the initial composition for any target $(\rho, \text{ratio}_{\text{C/O}}, Y_e)$ point in the parameter space.

The choice to introduce the spectator species into the **fuel** is deliberate: what governs the flame physics is the electron fraction of the unburned material ahead of the front, not that of the ash left behind it, since the ash is downstream and already reacted. An inert species keeps the fuel Y_e fixed at its target value by construction, independent of what happens to the composition once it crosses the flame front.

4.3.3 Comparison with Ne22

Ne22 ($Z/A \approx 0.4545$) was considered as a more physically motivated alternative, being a real, stable isotope. However, reaching a given Y_e shift with Ne22 requires a substantially larger mass fraction than with Ne40: for example, $Y_e = 0.49$ requires $X_s(\text{Ne22}) \approx 0.26$, compared to $X_s(\text{Ne40}) \approx 0.04$ from Equation (4.3.2), a difference of roughly a factor of six. Since the goal is to shift Y_e while perturbing the C-O fuel composition as little as possible, Ne40 was retained despite not corresponding to a physically real neon isotope in our stellar matter nuclear networks.

4.3.4 Box Sizing for $Y_e < 0.5$

The self-similar box-sizing strategy of Section 4.2.1 (Equation (4.2.1)) is defined in terms of the flame speed and width predicted by (Timmes & Woosley, 1992), which does not itself depend on Y_e . To extend this scaling to $Y_e < 0.5$, the (Schwab et al., 2020) (eq. 2) correction factor is applied multiplicatively,

$$\text{factor}(Y_e) = 1 + 96.8(0.5 - Y_e) \quad (4.3.4)$$

giving a corrected expected speed and flame width,

$$v_{\text{cond}}(\rho, \text{ratio}_{\text{C/O}}, Y_e) = v_{\text{cond, Timmes}}(\rho, \text{ratio}_{\text{C/O}}) \times \text{factor}(Y_e), \quad (4.3.5)$$

$$\ell(\rho, \text{ratio}_{\text{C/O}}, Y_e) = \ell_{\text{Timmes}}(\rho, \text{ratio}_{\text{C/O}}) / \text{factor}(Y_e), \quad (4.3.6)$$

both of which are substituted into Equation (4.2.1) in place of their Timmes-only counterparts when generating simulations away from $Y_e = 0.5$.

4.3.5 Limitations

Two approximations underlie this methodology and are worth stating explicitly. First, Ne40 is a numerical device rather than a physical isotope: no neon nuclei has $Z/A = 0.25$. Second, the correction factor of Equation (4.3.4) was calibrated by (Schwab et al., 2020) on O-Ne mixtures, not C-O; its use here is a working assumption to be tested against simulation results (Section 4.6), not

an established result for this composition. Some support for the underlying approach comes from (Schwab et al., 2020) themselves, who find the Y_e dependence of the flame speed to be largely insensitive to which neutron-rich species carries the excess neutrons in their models (their Fig. 7 gives consistent results for Ne22, Ne23, and Ne24), suggesting that the identity of the spectator species matters less than its mass fraction. On top of this, the parameters calculated from these adjusted values have no effect on the physical behavior and asymptotic speed of the flame, as shown in section 4.2.4, the only impact is in the simulated duration and the scope of the study, not on the physical level.

4.3.6 Parameter Grid

The full phase-space is:

$$\rho \in \{0.01, 0.05, 0.1, 0.2, 0.5, 1, 2, 4, 6, 8, 10\} \times 10^9 \text{g/cm}^3, \quad (4.3.7)$$

$$\text{ratio}_{\text{C/O}} \in \{0.2, 0.4, 0.5, 0.6, 0.8, 1.0\}, \quad (4.3.8)$$

$$Y_e \in \{0.5, 0.495, 0.490, 0.485, \dots\}, \quad (4.3.9)$$

with $Y_e = 0.5$ run first in every case to validate against (Timmes & Woosley, 1992) before extending to $Y_e \neq 0.5$. At the time of writing, the $Y_e = 0.5$ sweep has been completed (Section 4.6); the extension to variable Y_e is unfinished work due to time constraints (Section 4.7).

4.4 Development of the Data Post-Processing Method

4.4.1 Pipeline Architecture

The full analysis chain runs in four stages: `generate_simulations.py` sizes and submits a simulation following the self-similar scheme of Section 4.2.1, `Phlegethon` runs on the Genoa cluster and writes RPROFS snapshots, a flame-speed extraction stage processes these snapshots into a single speed estimate per simulation, and `analyze_results.py` aggregates the results across the parameter space into cached csv tables and comparison plots.

4.4.2 Flame Front Detection

The position of the flame front within a snapshot is estimated from two independent indicators and averaged: the radius at which the temperature profile crosses the midpoint between the fuel and ash temperatures, $(T_{\text{fuel}} + T_{\text{ashes}})/2$, and the radius of the peak nuclear energy generation rate $\dot{\epsilon}_{\text{nuc}}$. The mean temperature value crossing criterion replaced an earlier implementation based on the inflection point of $T(r)$, which proved erratic on profiles with local noise, since it relies on a well-behaved second derivative; the midpoint crossing only requires the fuel and ash plateaus themselves to be identifiable, which is a substantially weaker requirement, and as such, more consistent.

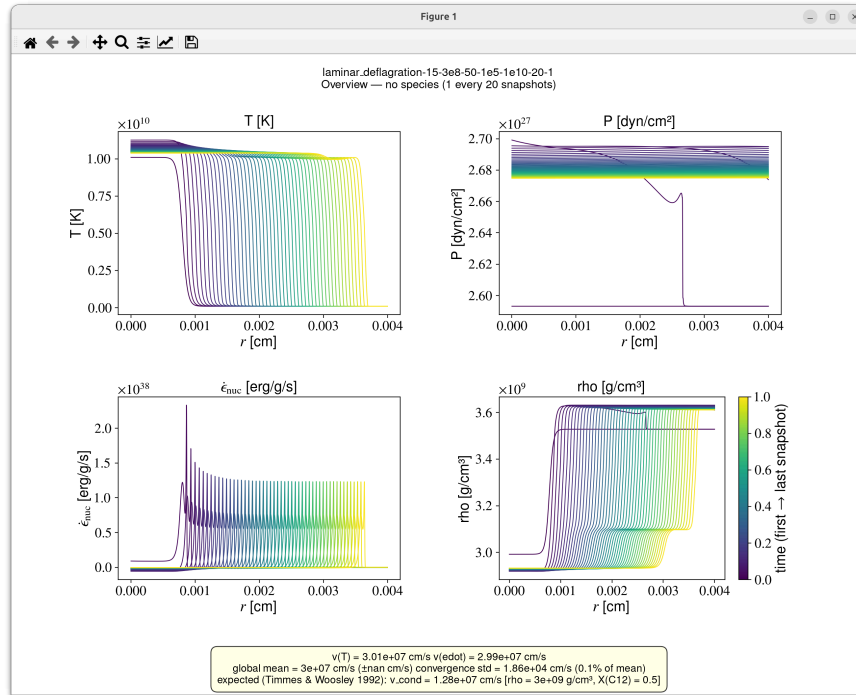


Figure 4.4.6: A case where the temperature briefly exceeds the stable ash temperature ahead of the true front, causing the inflection-point criterion to misidentify the front position.

In both cases, sub-cell resolution is recovered via a parabolic interpolation across the three cells adjacent to the remarkable point (mean temperature crossing or peak $\dot{\epsilon}_{\text{nuc}}$), rather than reporting the position at native grid resolution. Snapshots past the point where the flame reaches 90% of the box length are discarded, to exclude the influence of the domain boundary described in Section 4.2.5.

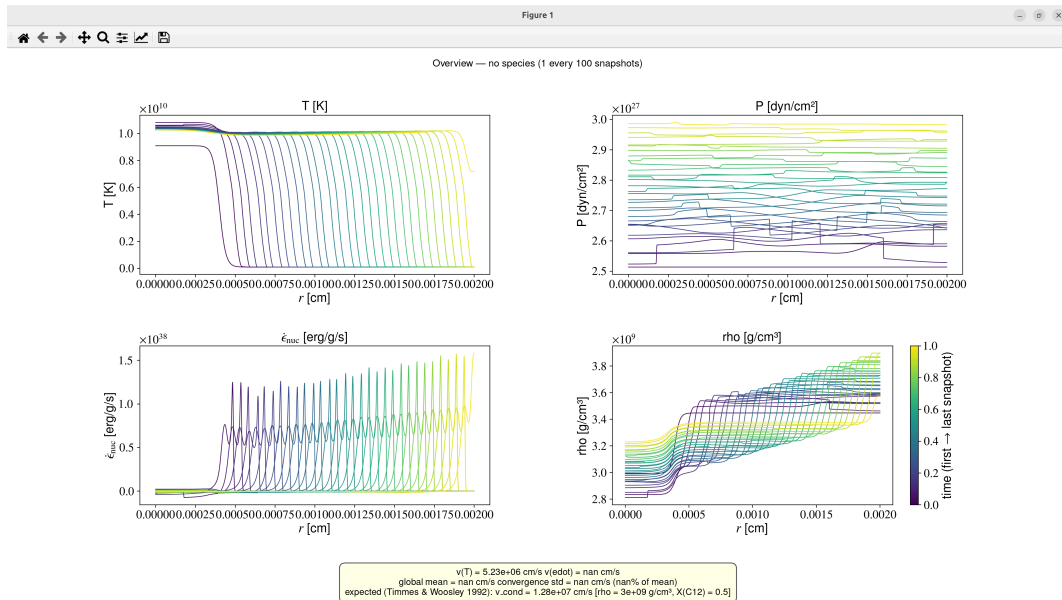


Figure 4.4.7: Radial profiles as the flame approaches the right-hand domain boundary: the pressure and density profiles show a visible upturn near the edge, motivating the 90% cutoff. This run also used a reduced ignition temperature without ignition issues.

4.4.3 Velocity Extraction

Three successive methods were used over the course of the project to convert a front-position time series $r(t)$ into a single flame speed estimate, each addressing a shortcoming of the previous one; the difficulties that motivated each transition are discussed in detail in Section 4.5.

Time-averaged velocity. The initial approach averaged the instantaneous velocity dr/dt over the full run. This estimator is biased whenever the transient discussed in Section 4.2.4 has not fully decayed by $t_{\max}$, which is the common case.

Polynomial fit. A low-order polynomial fit to $r(t)$, differentiated analytically, was adopted next to obtain a smooth velocity estimate without differencing noisy raw snapshots. It remains a numerically robust fallback but has no physical basis for extrapolating to the true asymptotic speed.

Integrated exponential relaxation fit. The current default method fits $r(t)$ directly with the physically motivated model

$$r(t) = r_0 + v_\infty t + A\tau(1 - e^{-t/\tau}) \quad (4.4.1)$$

so that v_∞ is a direct estimate of the asymptotic flame speed and τ of the transient's relaxation time. This functional form is motivated by the classical theory of traveling-wave solutions to reaction-diffusion systems: near a stable traveling wave, Fife & McLeod (1977) show that small perturbations decay exponentially in time, consistent with the flame speed acting as a stable physical attractor, in the sense of the eigenvalue flame-speed formalism introduced by Zel'dovich & Frank-Kamenetskii (1938), also underlying the formulation of Timmes & Woosley (1992). It should be stressed that both of these classical results were derived for comparatively simple reaction-diffusion systems, typically a single chemical reaction-progress variable without degenerate thermodynamics, fully compressible hydrodynamics, or a stiff multi-species nuclear reaction network. Here they are used as physical motivation for the functional form rather than as a rigorous derivation for this specific, considerably more complex multiphysics system like the ones we consider in this study. The superposition of several coupled effects, thermal conduction, compressible hydrodynamics, a 35-species reaction network, and, as discussed in Section 4.2.5, spurious acoustic reflections, gives additional mechanisms and timescales that a single exponential mode is not guaranteed to capture. This justification is therefore asymptotic and qualitative, and does not guarantee a single exponential mode dominates the transient from $t = 0$ onwards, a limitation revisited in Section 4.5 together with the fit-stability issues it produces in practice.

4.4.4 Robust Cross-Check

To guard against the fit-stability issues discussed in Section 4.5, every simulation is additionally processed by an independent, non-parametric estimator that never differentiates the raw position data directly:

- **Sliding-window local slopes.** $r(t)$ is split into consecutive windows, each fit by ordinary least squares to obtain a local velocity and its formal uncertainty.

- **Aitken Δ^2 extrapolation.** Aitken’s delta-squared method is applied to consecutive triplets of these local velocities to extrapolate the asymptotic value, with linear uncertainty propagation and rejection of triplets whose denominator is not resolved above its own noise floor. The method itself is detailed in Appendix B.1.
- **Late-window plateau.** An inverse-variance-weighted mean of the local velocities over the last fraction of the run serves as a numerically bullet-proof fallback with no extrapolation.

The three estimates, together with the exponential fit of Equation (4.4.1), are combined into a single inverse-variance-weighted robust estimate v_{robust} , flagged as agree when the individual estimates lie within 5% of one another and as disagreeing otherwise. A collapse guard compares the exponential fit’s v_{∞} against the run’s overall secant slope $(r(t_{\text{max}}) - r(t_0))/(t_{\text{max}} - t_0)$; if the two differ by more than a factor of 20 (or the fit gives a non-positive speed), the exponential fit is discarded in favor of v_{robust} .

4.4.5 Diagnostic and Caching Infrastructure

Because processing every simulation’s RPROFS snapshots from scratch is costly, `analyze_results.py` caches results in a CSV file, re-used on subsequent runs unless explicitly forced to recompute. Alongside the velocity estimates, each entry records the number of valid snapshots retained after the boundary cutoff, the total simulated time span T_{span} , and the ratio τ/T_{span} : a value exceeding unity flags a simulation whose transient likely has not been fully resolved within t_{max} , independent of whether the fit itself converged cleanly. These diagnostics are used throughout Section 4.6 to distinguish genuine physical discrepancies from simulations that would require a longer run.

Empirically, this ratio does not follow a simple fixed power-law dependence on speed: t_{max} reaches around 1.5τ at high density and high speed, but drops to only around 0.1τ at low density and low speed, with the divergence between the two accelerating rather than following a constant scaling exponent. This behavior, and its consequences for low-density simulations specifically, are examined in detail in Section 4.5.

4.5 Physical Challenges and Diagnostic of Transient Regimes

4.5.1 Long Transients and the Failure of Time-Averaging

The first systematic validation attempt ran 66 simulations at $Y_e = 0.5$ across the density and C/O grid of Section 4.3, to be compared against Timmes & Woosley (1992). The results were not conclusive: the instantaneous flame speed decreased monotonically over the full duration of most runs, with no clear plateau reached by t_{max} . Averaging the speed over the run, the initial approach described in Section 4.4, is therefore biased, and biased by an amount that depends on how much of the underlying transient happens to be covered by t_{max} , itself a function of $(\rho, \text{ratio}_{\text{C/O}}, Y_e)$ through the self-similar sizing of Section 4.2.1.

This inconsistency across the parameter space, rather than any single simulation looking obviously wrong, was the first indication that the extraction method itself, not only the underlying

physics, needed to be reconsidered, motivating the move to the exponential relaxation fit of Equation (4.4.1).

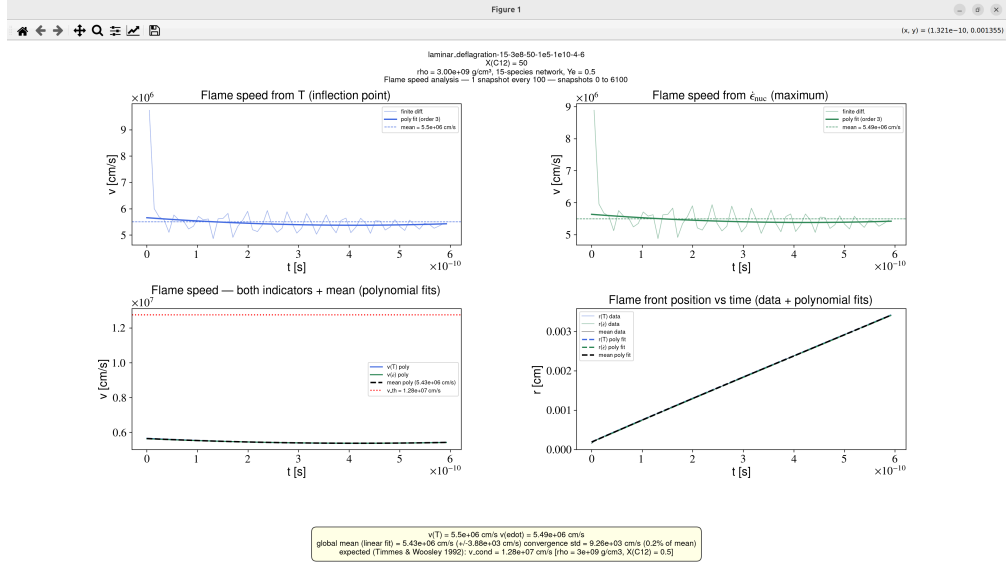


Figure 4.5.8: Polynomial-fit extraction on a well-behaved run: smooth and numerically stable, but with no physical basis for the extrapolated asymptotic value.

4.5.2 Exponential Fit Instability and the Collapse Guard

The nonlinear fit of Equation (4.4.1) can converge to a degenerate solution in which v_{∞} collapses toward zero while A and τ absorb the front's actual motion into the transient term. Diagnostic plots produced during the project showed this manifesting as fitted speeds many orders of magnitude below the expected value, in one case around 10^{-15} cm/s, without the fit raising any error.

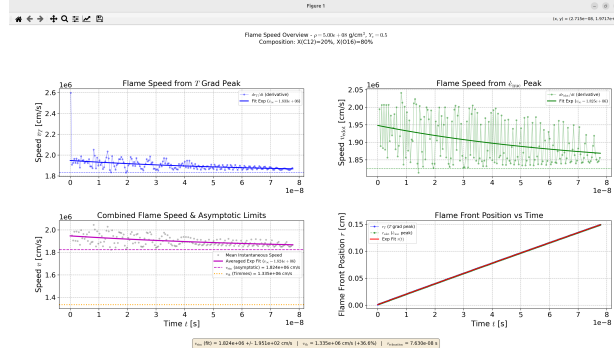


Figure 4.5.9: A well-conditioned fit: τ is well resolved within t_{max} .

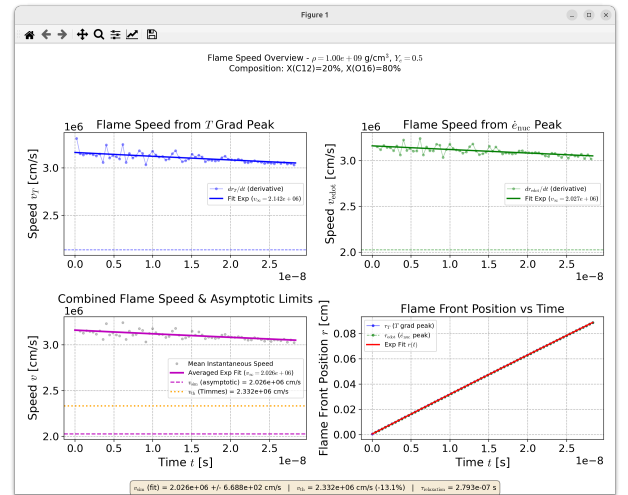


Figure 4.5.10: A degenerate fit: the extrapolated asymptote lies far outside the range the data actually cover.

The underlying cause, confirmed once the diagnostics of Section 4.4 were in place, is a strongly ill-conditioned fit: when t_{max} does not cover several τ , the three parameters v_{∞} , A , and τ become highly correlated, and the least-squares solver can settle on a degenerate combination that repro-

duces the observed $r(t)$ equally well numerically while being physically meaningless. As discussed in Section 4.5.5, this is precisely the regime that low-density simulations tend to fall into. The collapse guard and robust cross-check of Section 4.4 were introduced specifically to catch and correct for this failure mode rather than to prevent it outright, since the underlying data limitation, insufficient $t_{\max}$ relative to τ , cannot be fixed after the fact.

4.5.3 Sensitivity to the Extraction Method

A further complication surfaced when comparing extraction methods directly against one another: applying Aitken Δ^2 extrapolation to the raw, undifferenced velocity signal produced more coherent results on some simulations but substantially more aberrant ones on others, such that the overall trend across the parameter space was noisier than with the simpler polynomial fit it was meant to replace. The reason is that Aitken extrapolation is effectively a second-difference operator, and second differences strongly amplify high-frequency noise, of the kind discussed in Section 4.5.6, when applied directly to already-noisy raw data. This was resolved not by abandoning Aitken extrapolation but by applying it to smoothed, sliding-window local slopes rather than to the raw signal, and by cross-checking the result against independent estimators, exactly the robust cross-check pipeline described in Section 4.4.

4.5.4 The Y_e Labeling Bug

The most consequential issue encountered was not a numerical or physical effect at all. For an extended period, comparison plots at fixed ρ and composition showed large, apparently random jumps in flame speed between adjacent density points of the same C/O ratio, jumps too large and too irregular to be explained by the transient and fit issues above. The eventual diagnosis was that `read_sim_params()`, the routine responsible for reading each simulation's parameters back from its configuration files, never actually recomputed Y_e from the composition on disk: it silently returned a hardcoded default of 0.5 for every simulation, regardless of the true value set through the Ne40 spectator fraction of Section 4.3. As a direct consequence, any Y_e filter matched every simulation unconditionally, and comparison plots ostensibly restricted to $Y_e = 0.5$ silently mixed in simulations run at $Y_e = 0.495, 0.490$, and other values, since nothing in the pipeline could actually tell them apart. Once identified, the fix was to recompute Y_e directly from the Ne40 mass fraction present in each simulation's own composition, following Equation (4.3.2) in reverse. This episode is worth recording explicitly: what appeared for some time to be a severe physical or numerical inconsistency in the flame-speed data was, in large part, a data-integrity bug in the post-processing pipeline itself, underscoring the need to validate the diagnostic tools thoroughly before drawing physical conclusions from their output.

4.5.5 The τ -Speed Relation and Low-Density Simulations

The diagnostic ratio τ/T_{span} introduced in Section 4.4 revealed a systematic pattern across the parameter space: at high density and high speed, $t_{\max}$ (sized from the expected Timmes speed, Section 4.2.1) reaches around 1.5τ , while at low density and low speed it drops to only around 0.1τ , a difference of more than an order of magnitude in the ratio itself. Notably, this progression

does not appear linear: the growth of τ relative to $t_{\max}$ accelerates as density decreases, rather than following a fixed power-law dependence on speed. A naive argument, that both $t_{\max}$ and τ should scale as ℓ/v , with the flame width ℓ itself inversely proportional to v (Section 4.2.1), would predict $t_{\max} \propto 1/v^2$ for both quantities and hence a roughly **constant** ratio between them, which is directly contradicted by the accelerating divergence actually observed. This suggests that τ picks up an additional density dependence beyond the simple flame-width-crossing argument, possibly related to the same low-density regime change flagged by the ignition-temperature issue of Section 4.2.4, though this connection has not been established. A precise characterization of this behavior has not been carried out and is left as an avenue for future work (Section 4.7). In practice, the consequence is direct: low-density simulations are the ones most starved of resolved relaxation time relative to τ , making them the most likely to trigger the collapse guard of Section 4.4 and to show the largest residual disagreement between extraction methods in the results of Section 4.6.

4.5.6 Acoustic Reflections: Consequences for the Extracted Signal

The physical origin of this oscillatory noise, a compression wave seeded by the artificial ignition energy and modulating the local density ahead of the flame front on each pass, was established in Section 4.2.5. Its consequence for the extraction methods of Section 4.4 is that this noise is superposed on the smoother relaxation transient discussed above, rather than replacing it: any fitting method has to contend with both simultaneously. This is part of why the sliding-window averaging step of the robust cross-check, rather than any single global fit, proved necessary: averaging over a window comparable to or larger than the reflection period suppresses this oscillatory component before the extrapolation step is applied, whereas a fit applied directly to the raw signal has no such protection.

4.5.7 Summary

Taken together, these five issues, biased time-averaging, fit degeneracy, extraction-method noise sensitivity, the Y_e labeling bug, and the density-dependent transient length, compounded rather than occurring in isolation, and disentangling them required building the diagnostic infrastructure of Section 4.4 alongside the physical investigation itself. The main lesson carried forward is that the reliability of a derived physical quantity, here the flame speed, cannot be assessed from the quantity alone; it required cross-checking independent extraction methods against each other and validating the labeling of the data feeding into them, which is what ultimately exposed the Y_e bug that a purely physics-focused investigation would not have caught.

4.6 Results and Analysis

4.6.1 Validation Against Timmes and Woosley (1992) at $Y_e = 0.5$

The full $Y_e = 0.5$ sweep of Section 4.3, 66 simulations spanning eleven densities and six C/O ratios, has been completed and processed through the pipeline of Section 4.4. Figure 4.6.11 shows the resulting flame speed, v_{robust} where the collapse guard flagged the exponential fit as unreliable and v_{sim} otherwise, against the prediction of Timmes & Woosley (1992) for each composition.

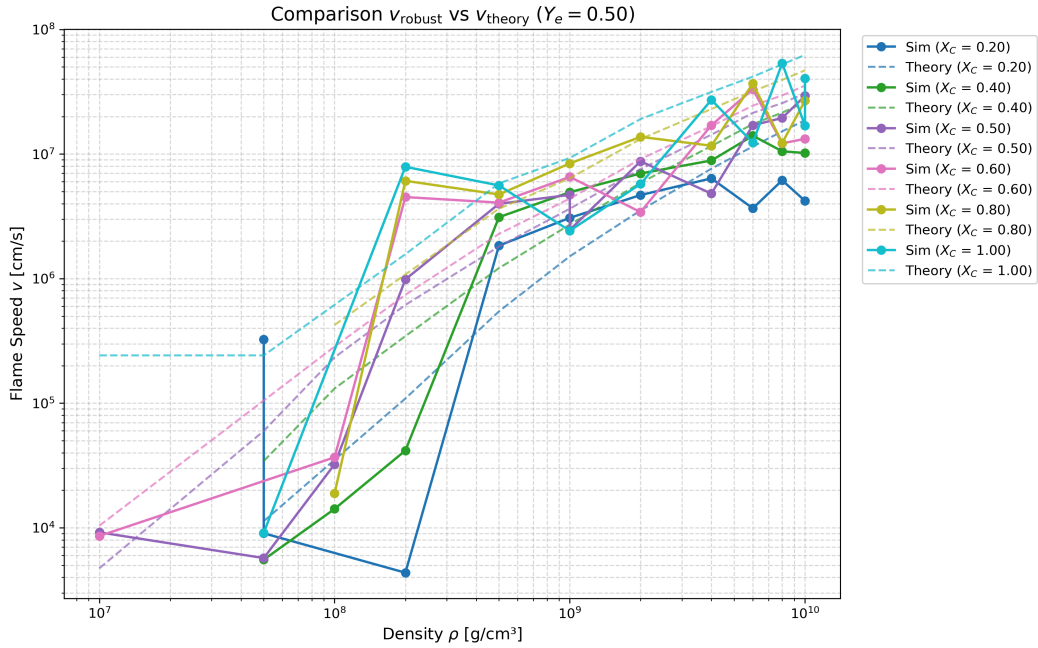


Figure 4.6.11: Simulated flame speed vs. the prediction of Timmes & Woosley (1992) at $Y_e = 0.5$, across the full density and C/O grid of Section 4.3, with exponential fitting.

At $\rho \gtrsim 5 \times 10^8 \text{ g/cm}^3$, Figure 4.6.12 (polynomial fit) shows a clean, monotonic trend in log-log space for every composition, correctly ordered relative to one another, though offset from the Timmes & Woosley (1992) prediction by a moderate but fairly consistent factor. This confirms that the underlying physics behaves consistently across the composition grid in this density range, even though the polynomial estimator itself is not expected to converge to the true asymptotic speed (Section 4.4).

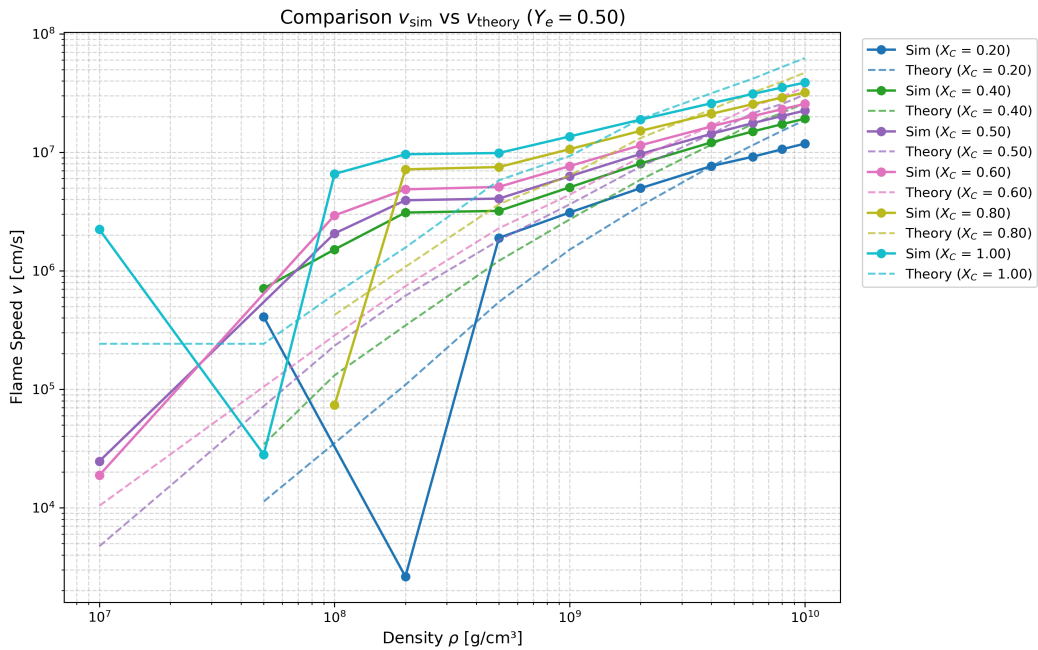


Figure 4.6.12: The same $Y_e = 0.5$ validation, extracted with the polynomial-fit method rather than the exponential relaxation fit of Figure 4.6.11.

Figure 4.6.11 (exponential fit) tells a more mixed story over the same density range. On the subset of simulations where the fit is well-conditioned (Section 4.5.5), the extracted speed sits closer to the Timmes & Woosley (1992) prediction than the polynomial estimate on the same points, consistent with the exponential model’s stronger physical motivation (Section 4.4). However, since a substantial fraction of simulations in this range still fail to produce a well-conditioned fit, points from different compositions become interleaved rather than staying cleanly separated: some higher-density points fall below lower-density ones of a different composition, an ordering violation not observed in the polynomial extraction at these densities specifically. At lower densities, both extraction methods show outliers, and the polynomial fit’s advantage in this respect is confined to the higher-density regime. In aggregate, the exponential fit is therefore more accurate on the individual simulations where it succeeds, but less reliable as a global picture, since a failed fit is not visually distinguishable from a successful one without consulting the diagnostics of Section 4.4. Below $\rho \approx 5 \times 10^8 \text{ g/cm}^3$, several points depart from the theoretical curve by multiple orders of magnitude, in some cases dropping to values that are clearly non-physical.

4.6.2 Diagnosis of the Low-Density Discrepancy

The low-density outliers in Figure 4.6.11 are not scattered randomly across the grid: they recur at the same handful of density values across several different C/O ratios simultaneously. This pattern, several compositions failing together at the same ρ rather than isolated simulations failing independently, points to a systematic cause tied to density itself rather than to per-simulation numerical noise.

Cross-referencing these outliers against the diagnostics of Section 4.4 confirms this: the affected simulations are overwhelmingly the same ones flagged by a high τ/T_{span} ratio and by disagreement (agree = false) between the independent extraction methods of Section 4.4, consistent with the accelerating growth of τ at low density identified in Section 4.5. In other words, the discrepancy visible in Figure 4.6.11 is best explained, at the current stage, as t_{max} being insufficient to resolve the transient at low density, rather than as a genuine physical disagreement with Timmes & Woosley (1992). This does not rule out a real physical effect being present as well, but it cannot currently be distinguished from the extraction issue with the available run lengths.

4.6.3 Phase-Space Maps

A two-dimensional map and a three-dimensional surface of $v(\rho, \text{ratio}_{\text{C/O}})$ at $Y_e = 0.5$ were produced using the tooling described in Section 4.4, but are not yet conclusive: the low-density artifacts of the previous section propagate directly into these visualizations and currently dominate their appearance at the low- ρ end of the grid.

4.6.4 Current Status

At the time of writing, the $Y_e = 0.5$ validation is complete for $\rho \gtrsim 5 \times 10^8 \text{ g/cm}^3$, with the low-density regime requiring the further investigation outlined in Section 4.5.5 before it can be considered resolved. The extension of the parameter sweep to $Y_e \neq 0.5$ (Section 4.3) and the

derivation of a combined fitting formula $v_l = f(\rho, X_i, Y_e)$ remain future work, discussed in Section 4.7.

4.7 Project Planning and Chronological Overview

Unlike an industrial project with a fixed upfront schedule, this internship's planning was inherently non-linear: numerical anomalies and unexpected physical behavior repeatedly opened new investigation branches that could not have been scheduled in advance (see also Section 2.3). The chronology below reconstructs the actual sequence of work rather than a plan set out at the start.

Late May, Onboarding. Initial setup of the working environment, cluster access, and first familiarization with Phlegethon's runtime parameters, supported by the onboarding guidance described in Section 2.3. A dedicated 1D Cartesian geometry branch was created for the project.

June, Setup and preliminary numerical tests. The bulk of the work described in Section 4.2 took place in this period: progression of the nuclear network from 7 to 15 to 35 species, the self-similar box-sizing scheme, OpenMP scalability testing, and the convergence studies on grid resolution, box size, ignition temperature, and ignition spot size (Section 4.2.4). The spurious acoustic wave reflections (Section 4.2.5) were also identified and characterized during this period, alongside the design of the Ne40 spectator-species methodology for variable Y_e (Section 4.3).

July, Pipeline development and first validation attempt. The post-processing pipeline (Section 4.4) was built out from an initial naive time-averaged velocity to a polynomial-fit estimator, and the first systematic validation sweep of 66 simulations at $Y_e = 0.5$ was run against Timmes & Woosley (1992). The long transients described in Section 4.5.1 were identified from this sweep, motivating the move to the integrated exponential relaxation fit of Equation (4.4.1).

Late July to August, Diagnostic infrastructure and bug fixing. The exponential fit's stability issues (Section 4.5.2) and the sensitivity of extracted speeds to the choice of extraction method (Section 4.5.3) were diagnosed and addressed through the robust cross-check pipeline (Section 4.4.4): sliding-window regression, Aitken Δ^2 extrapolation, and the collapse guard. The Y_e labeling bug (Section 4.5.4) was identified and fixed during this period, after which the $Y_e = 0.5$ validation could be re-run and properly assessed (Section 4.6).

August, Results, diagnosis, and report writing. The current results were compiled and the low-density discrepancy diagnosed against the τ/T_{span} ratio (Section 4.6.2). Report writing proceeded concurrently with the remaining diagnostic work, using simulation runtime as an opportunity to draft sections not dependent on final results, following the guidance received during the internship's weekly SLH meetings (Section 2.3).

Not completed within the internship. The extension of the parameter sweep to $Y_e \neq 0.5$ (Section 4.3.6), resolution of the low-density transient issue, and derivation of the combined fitting formula $v_l = f(\rho, X_i, Y_e)$ remain future work, discussed in Section 5.1.

5 Conclusion and Perspectives

This project set out with three objectives (Section 3.4): deriving a fitting formula for the laminar deflagration flame speed $v_l = f(\rho, X_i, Y_e)$, validating Phlegethon against the existing literature on conductive flames, and documenting the resulting post-processing tools for reuse within the group.

The second and third objectives have largely been met. A self-similar simulation design (Section 4.2) makes it possible to sweep density and composition at a roughly constant computational cost per simulation, and the resulting flame speeds agree with Timmes & Woosley (1992) to within a factor of a few for $\rho \gtrsim 5 \times 10^8 \text{ g/cm}^3$ (Section 4.6). A complete post-processing pipeline was developed from scratch over the course of the internship: flame-front detection, an exponential relaxation fit motivated by the classical theory of traveling combustion waves, a robust, non-parametric cross-check against it, and a set of diagnostics (τ/T_{span} , agree) that flag unreliable extractions rather than silently returning them (Section 4.4). This tooling, together with the scripts used to generate and scan the simulation parameter space, is intended to remain usable by the PSO group beyond the scope of this internship.

The first objective, a combined fitting formula depending on all three of ρ , X_i , and Y_e , remains incomplete. Two things stand in its way at the time of writing. First, the low-density regime ($\rho \lesssim 5 \times 10^8 \text{ g/cm}^3$) is not yet resolved: Section 4.5 traced this to τ growing disproportionately fast relative to t_{max} as density decreases, a behavior confirmed empirically but not yet explained physically. Second, the Y_e sweep itself (Section 4.3) has not been carried out beyond $Y_e = 0.5$, so the Ne40 spectator methodology and the Schwab-derived box-scaling correction remain validated only in principle, not against simulation results.

Several of the difficulties encountered along the way were not primarily physical. The most significant, a labeling bug that silently defaulted every simulation's Y_e to 0.5 regardless of its true composition (Section 4.5.4), cost a substantial amount of time before being identified, and only came to light once results were cross-checked against multiple independent extraction methods rather than trusted individually. This turned out to be one of the more instructive parts of the project: for a numerical study of this kind, validating the post-processing pipeline itself is as necessary as validating the underlying physics, and the two are not always easy to tell apart from the output alone.

5.1 Perspectives

Several concrete directions follow directly from the state described above:

- **Resolving the low-density transient.** A precise characterization of the τ -density relation identified in Section 4.5.5, and correspondingly longer t_{max} values at low density, are needed before the validation of Section 4.6 can be considered complete across the full density range.

- **Completing the Y_e sweep.** Extending the simulations of Section 4.3 to $Y_e \neq 0.5$ would test the Ne40 spectator methodology and the Schwab-derived correction directly, rather than relying on the working assumptions of Section 4.3.5.
- **The combined fitting formula.** Once both of the above are addressed, fitting $v_{l(\rho, X_i, Y_e)}$ across the full three-parameter grid becomes possible, delivering on the project's original objective.
- **Larger nuclear networks.** The 56-species network remains non-functional, most likely due to the $O(n^3)$ scaling of the dense-matrix linear algebra used by the network solver as network size grows (Section 4.2.2). A sparse-matrix reformulation, discussed further in Appendix B, is a natural next step, and would also bring the 35-species results used throughout this work closer to the large-network results of Timmes & Woosley (1992).
- **Beyond 1D.** As noted in Section 3.3, this study was deliberately restricted to a planar 1D geometry. Extending the validated post-processing pipeline to multi-dimensional simulations would allow the flame speed formula, once derived, to be tested against curvature and Rayleigh-Taylor effects absent from the present setup.

Beyond the specific scientific results, this internship provided a first sustained experience with high-performance computational astrophysics: working with an actively developed research code, debugging a numerical pipeline end to end rather than a single script, and communicating results, including unresolved problems, to a research group on a weekly basis. The gap between a simulation producing a number and that number being trustworthy turned out to be the central, recurring theme of the project.

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Glossary / Glossaire

Acronym	Full Name	Definition
SN Ia	Type Ia supernova	Thermonuclear explosion of a carbon-oxygen white dwarf exceeding the Chandrasekhar mass limit in a binary system.
WD	White dwarf	Compact and degenerate stellar remnant of a low-to-intermediate-mass star. The maximum mass of a white dwarf is the Chandrasekhar limit ($\approx 1.4M_{\odot}$).
deflagration	Thermonuclear deflagration	Subsonic combustion regime sustained by thermal conduction coupled with nuclear fusion reactions.
EoS	Equation of state	Thermodynamic relation connecting pressure, mass density, and temperature of stellar matter.
network	Nuclear reaction network	Set of nuclear reactions and their rates that describe the evolution of stellar matter in thermonuclear fusion reactions.
flame speed	Laminar flame speed	Speed at which a flame propagates through a combustible mixture under idealized conditions in a laminar flow.
HPC	High-performance computing	Computing infrastructure using supercomputers or parallel processing clusters to run complex, computationally intensive simulations.
VPN	Virtual private network	Encrypted network connection that enables secure access to internal institute servers and remote resources.
SSH	Secure shell	Cryptographic network protocol used for secure remote command-line login and system administration on HPC clusters.

NSE	Nuclear statistical equilibrium	State reached at extremely high temperatures where forward and reverse nuclear reactions balance out, setting isotopic abundances strictly via thermodynamic state variables.
HEDP	High energy density physics	Study of matter under extreme conditions of pressure and density, relevant to both stellar interiors and inertial confinement fusion.
ICF	Inertial confinement fusion	A type of fusion energy research that uses high-power lasers or particle beams to compress and heat micro-capsules of thermonuclear fuel.
CE	Common envelope	Transient phase in binary star evolution where both stars orbit within a shared gaseous envelope, leading to rapid orbital decay and mass loss.
Phlegethon	Phlegethon	Open-source Fortran finite-volume fully compressible stellar hydrodynamics code, including thermonuclear reaction networks.
LTE	Local thermodynamic equilibrium	
ODE	Ordinary differential equation	
FV	Finite volume	
MP5	5th-order Monotonicity-Preserving scheme	
SSP-RK3	Strong Stability Preserving Runge-Kutta Order 3	
CFL	Courant-Friedrichs-Lewy	
STS	Super-Time-Stepping	
BE	Backward Euler	

MPI	Message Passing Interface	Standardized library specification for distributed-memory parallel programming across compute nodes.
OpenMP	OpenMP	API for shared-memory multi-threaded parallel programming within a single compute node.
JINA REACLIB	JINA REACLIB	Compiled database of thermonuclear reaction rates for astrophysical applications, maintained by the Joint Institute for Nuclear Astrophysics.
main sequence	Main sequence	Core hydrogen-burning phase of stellar evolution during which a star's luminosity and temperature are set primarily by its mass, occupying a well-defined band on the Hertzsprung-Russell diagram.

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A Additional Figures and Data

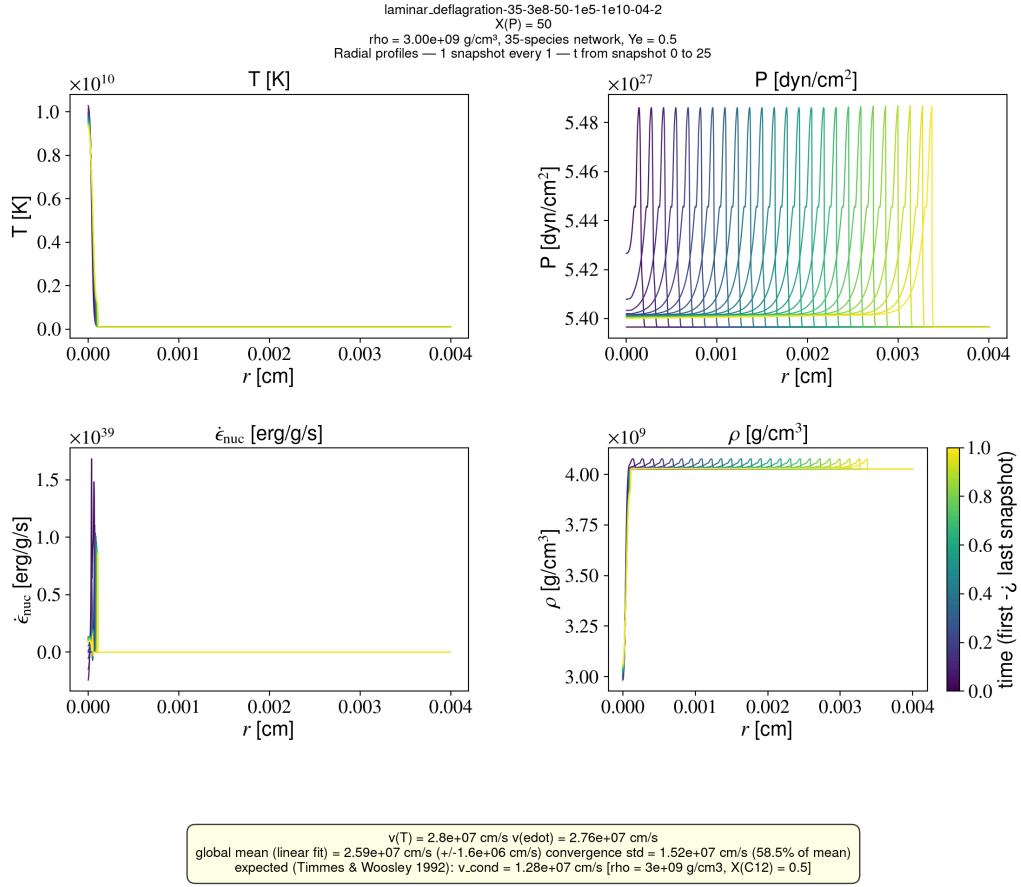


Figure 1.1.13: Radial profile overview from one of the OpenMP scalability test runs of Section 4.2.3 (15-species network, 10 000 timesteps).

A.1 Sensitivity Analyses to Initial Conditions and Limiting Regimes

A.1.1 Impact of Domain Size and Acoustic Oscillations

This series of figures illustrates the sensitivity of the flame-speed calculation to the dimensions of the numerical domain and to acoustic reflections.

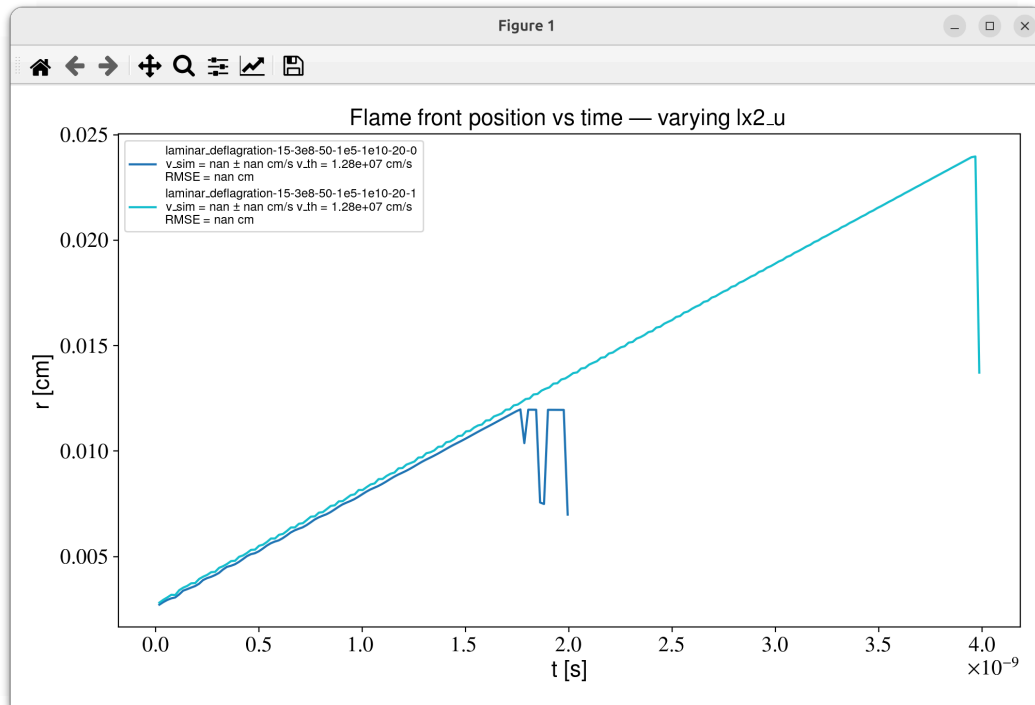


Figure 1.1.14: Evolution of the flame-front position for different box lengths ($x2u$). When the front reaches the right end of the domain, its interaction with the boundary condition causes a numerical divergence (nan). This justifies the analysis stopping criterion and data cutoff set at 90% of the domain length ($0.9L_x$) in the data post-processing.

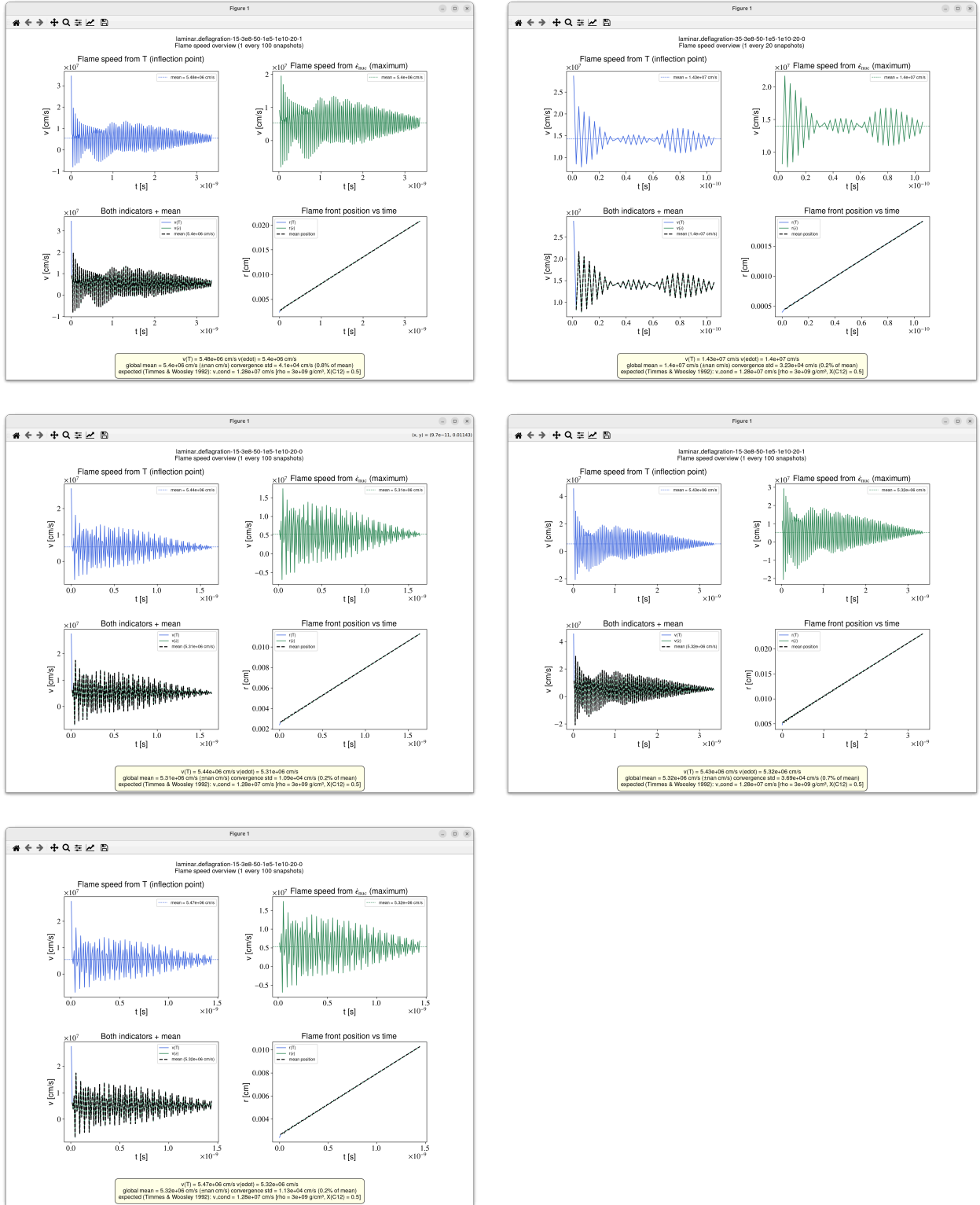


Figure 1.1.15: Examples of instantaneous velocity signals $v(t)$ (obtained from the inflection point of T and the maximum of ϵ_{nuc}) strongly affected by noise from compression waves. The oscillation frequency depends directly on the box length $x2u$, confirming the presence of acoustic waves in the box. The asymptotic time average makes it possible to extract the theoretical velocity despite these oscillations, although the signal remains highly noisy, which led to difficulties in drawing a conclusion for the project.

A.1.2 Spatio-Temporal Convergence Test (x_t)

This convergence study concerns the variation of x_t , the fraction of the box that is set to T at $t=0$ to initiate the deflagration, for a 15-species network ($\rho = 3 \times 10^9 (\text{g/cm})^3$, $X(\text{C12}) = 0.5$). The aim was to find the limiting value of x_t that injects the energy required for ignition without exceeding this value.

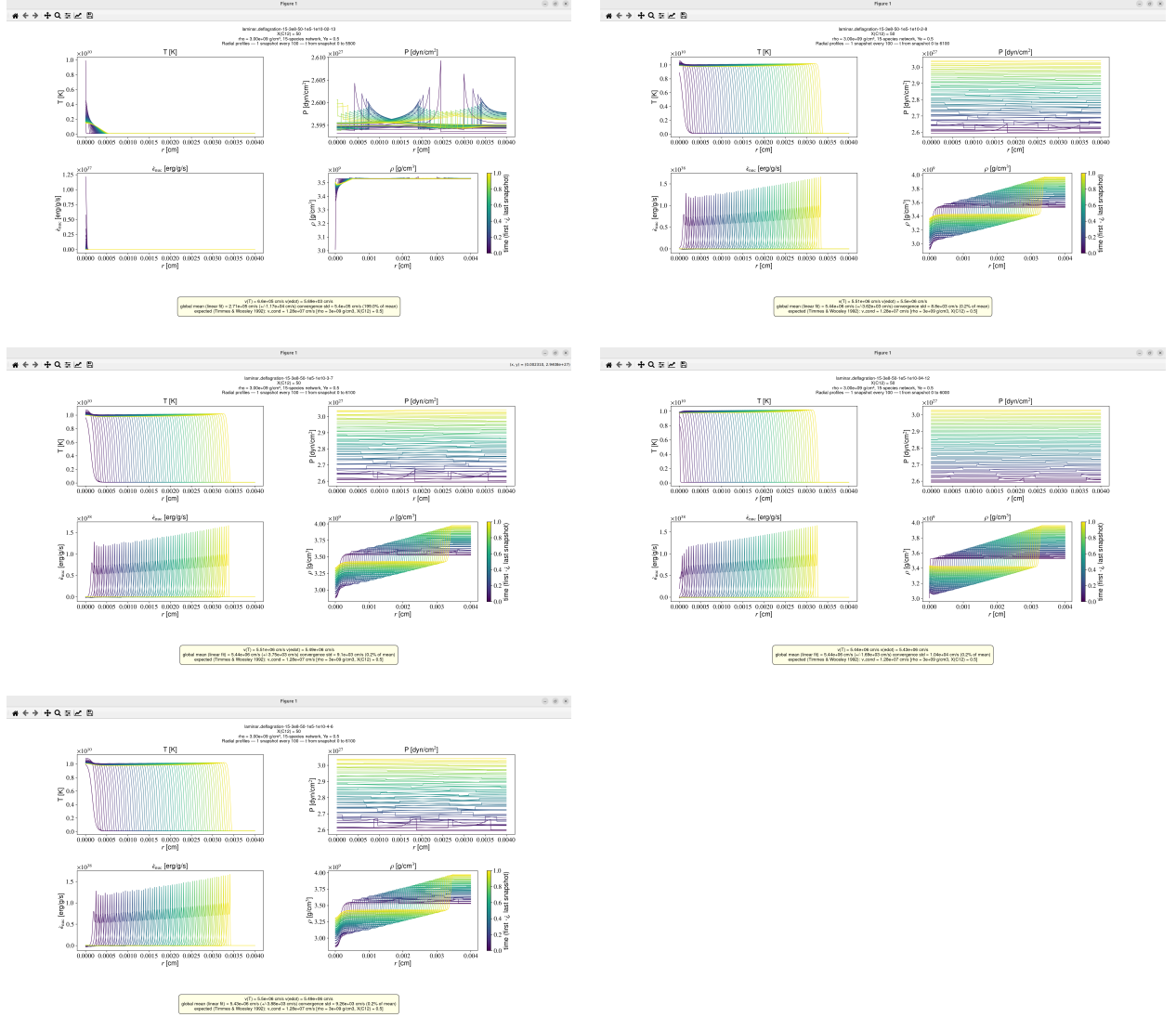


Figure 1.1.16: Spatio-temporal profiles (T , P , ϵ_{nuc} , ρ) for different values of the parameter x_t . For ignition regions that are too small ($x_t \leq 0.2$), the flame does not reach its steady state. Above this threshold, the deflagration is self-sustained. There is therefore a limiting value.

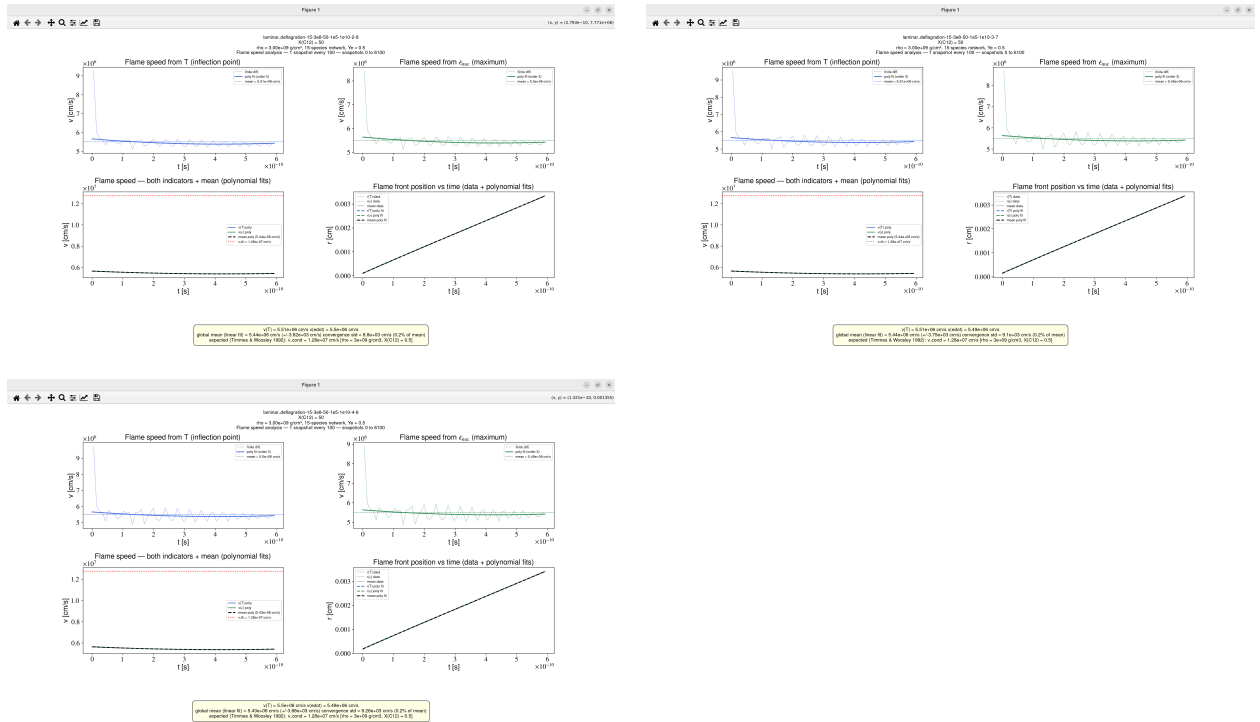


Figure 1.1.17: The same results are obtained for the speed_overview velocity curves.

A.1.3 Extinction Regime by Diffusion and Out-of-Domain Photodisintegration

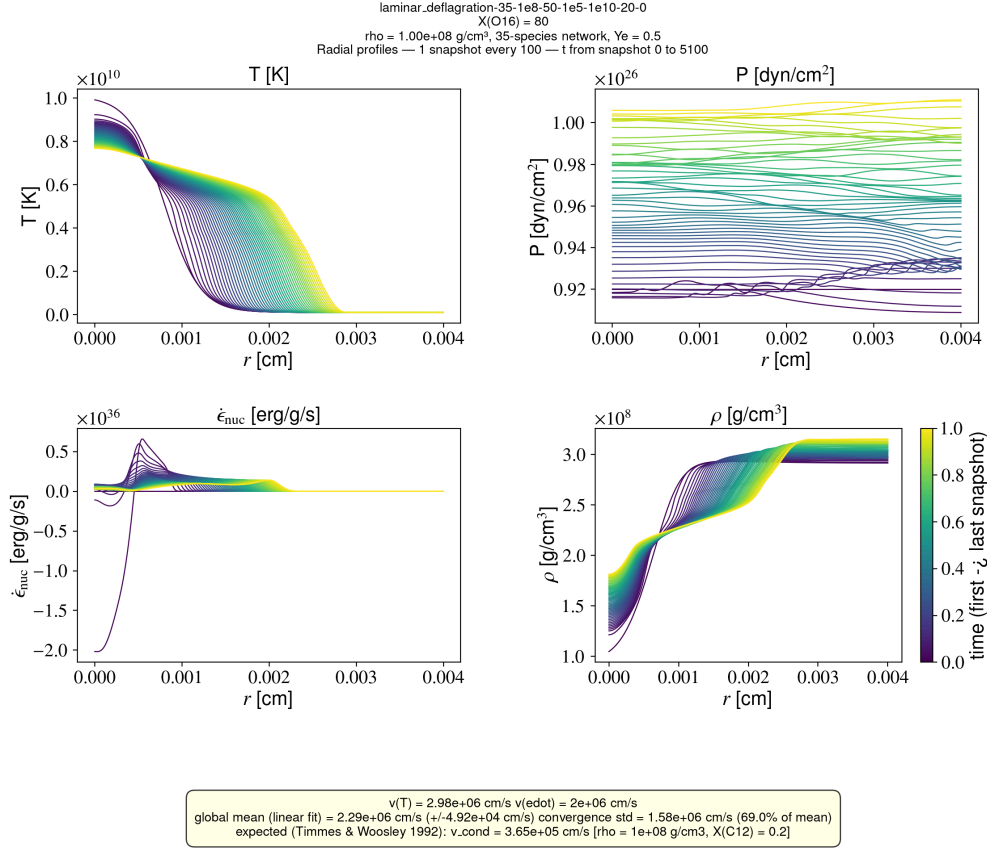


Figure 1.1.18: Spatio-temporal profiles of a simulation in a quasi-ignition regime followed by extinction through thermal diffusion ($X_{\text{O16}} = 0.8$, $\rho = 10^8 \text{ g/cm}^3$, 35-species network). An extremely large negative value of nuclear energy generation ($\dot{\epsilon}_{\text{nuc}} < -2 \times 10^{36} \text{ erg/g/s}$) is observed at the very beginning of the simulation. This massive energy absorption corresponds to intense photodisintegration, amplified by exploring a temperature/density range outside the validity limits of the JINA REACLIB reaction-rate library.

B Detailed Numerical Methods

This appendix expands on three methods summarized in the main text: the Aitken Δ^2 extrapolation of Section 4.4.4, the HLLC Riemann solver of Section 4.1.2, and the explicit scalar form of the conservation system introduced in vector notation in Section 4.1.1.

B.1 Aitken Δ^2 Extrapolation

B.1.1 Derivation

Consider a sequence of local velocity estimates $v_1, v_2, v_3, \dots$, obtained from the sliding-window regression of Section 4.4.4, and assume it relaxes geometrically toward an asymptote,

$$v_i = v_\infty + Ar^i \quad (2.1.1)$$

for some ratio r with $|r| < 1$ and amplitude A . This is the discrete analogue of the continuous exponential relaxation model of Equation (4.4.1), sampled at the window centers. Three consecutive terms v_1, v_2, v_3 give three equations in the three unknowns v_∞, A , and r , which can be solved for v_∞ alone. Writing the first and second differences $\Delta v_1 = v_2 - v_1$ and $\Delta^2 v_1 = v_3 - 2v_2 + v_1$, the geometric ansatz gives $\Delta v_1 = Ar(r-1)$ and $\Delta^2 v_1 = Ar(r-1)^2$, so that

$$v_\infty = v_1 - \frac{(\Delta v_1)^2}{\Delta^2} v_1 = \frac{v_3 v_1 - v_2^2}{v_3 + v_1 - 2v_2} \quad (2.1.2)$$

which is Aitken's Δ^2 formula. Unlike a direct nonlinear least-squares fit of the full sequence to the exponential model, Equation (2.1.2) requires only algebraic operations on three numbers, and cannot itself diverge to a local minimum the way a multi-parameter nonlinear solver can.

B.1.2 Uncertainty Propagation and Ill-Conditioning

Each v_i carries a formal uncertainty σ_i from the underlying window regression. Treating Equation (2.1.2) as a function of (v_1, v_2, v_3) , linear error propagation gives

$$\sigma_{v_\infty}^2 = \sum_{i=1}^3 \left(\frac{\partial v_\infty}{\partial v_i} \right)^2 \sigma_i^2, \quad (2.1.3)$$

with partial derivatives, writing $D = v_3 + v_1 - 2v_2$ for the denominator of Equation (2.1.2),

$$\frac{\partial v_\infty}{\partial v_1} = \frac{v_3 - v_\infty}{D}, \quad \frac{\partial v_\infty}{\partial v_2} = \frac{2v_\infty - 2v_2}{D}, \quad \frac{\partial v_\infty}{\partial v_3} = \frac{v_1 - v_\infty}{D}. \quad (2.1.4)$$

Because D appears in the denominator of every partial derivative, Equation (2.1.2) is intrinsically ill-conditioned whenever D is small: this is precisely the regime in which the sequence has barely started to curve, i.e. is close to linear over the three sampled points, and Aitken extrapolation amounts to dividing two small, noisy quantities. A triplet is treated as well-conditioned only when D is resolved above its own propagated uncertainty $\sigma_D = \sqrt{\sigma_1^2 + \sigma_3^2 + 4\sigma_2^2}$, specifically $|D| > 2\sigma_D$; triplets failing this criterion are excluded from the combined estimate (Section 4.4.4) rather than included with an artificially small formal uncertainty.

B.2 HLLC Riemann Solver

The HLLC flux of Equation (4.1.13) (Section 4.1.2) requires estimates of the left, right, and contact wave speeds S_L, S_R, S_* , following Toro et al. (1994). Given left and right states U_L, U_R with density ρ_K , normal velocity u_K , pressure P_K , and sound speed c_K ($K \in \{L, R\}$), the wave-speed estimates are

$$S_L = \min(u_L - c_L, u_R - c_R), \quad S_R = \max(u_L + c_L, u_R + c_R), \quad (2.2.1)$$

$$S_* = \frac{P_R - P_L + \rho_L u_L (S_L - u_L) - \rho_R u_R (S_R - u_R)}{\rho_L (S_L - u_L) - \rho_R (S_R - u_R)}. \quad (2.2.2)$$

The star-region states U_{*K} used in Equation (4.1.13) are then

$$U_{*K} = \rho_K \left(\frac{S_K - u_K}{S_K - S_*} \right) \left(\begin{array}{c} 1 \\ S_* \\ \frac{E_K}{\rho_K} + (S_* - u_K) \left[S_* + \frac{P_K}{\rho_K(S_K - u_K)} \right] \end{array} \right) \quad (2.2.3)$$

for the density, normal momentum, and total energy components respectively, with the transverse momentum and species mass fraction components of U_{*K} carried over unchanged from U_K , since they are advected passively across the contact wave.

B.3 Explicit 1D Form of the Conservation System

Section 4.1.1 presents the governing equations in vector form (Equation (4.1.1)). Restricted to the 1D planar geometry used throughout this study ($\frac{\partial}{\partial y} = \frac{\partial}{\partial z} = 0$, single velocity component u), the system reduces to five coupled scalar equations, one continuity equation, one momentum equation, one energy equation, and $N_{\text{species}} - 1$ independent species equations (the last being fixed by the constraint $\sum_k X_k = 1$):

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0, \quad (2.3.1)$$

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2 + P)}{\partial x} = \rho g, \quad (2.3.2)$$

$$\frac{\partial E}{\partial t} + \frac{\partial((E + P)u)}{\partial x} = \rho u g + \frac{\partial}{\partial x} \left(K \frac{\partial T}{\partial x} \right) + \rho \dot{\epsilon}_{\text{nuc}}, \quad (2.3.3)$$

$$\frac{\partial(\rho X_k)}{\partial t} + \frac{\partial(\rho X_k u)}{\partial x} = \rho \dot{\omega}_k, \quad k = 1, \dots, N_{\text{species}} - 1. \quad (2.3.4)$$

These are the equations actually discretized cell-by-cell by the finite-volume scheme of Section 4.1.2 the vector notation of Section 4.1.1 is retained in the main text for compactness and to keep the formulation independent of the number of species and spatial dimensions used in a given run.